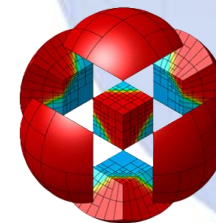
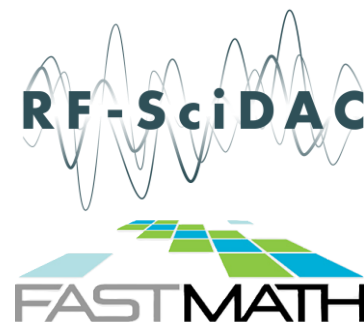


Updates in the framework for mixed systems of MFEM

MFEM Community Workshop
Sep 22 – 25, 2026

This material is based upon work supported by the U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research, Scientific Discovery through Advanced Computing (SciDAC) Program through the FASTMath Institute, under contract number DE-SC0021285, and RF-SciDAC Center, under contract number DE-SC0024369.

Prepared by LLNL under Contract DE-AC52-07NA27344.

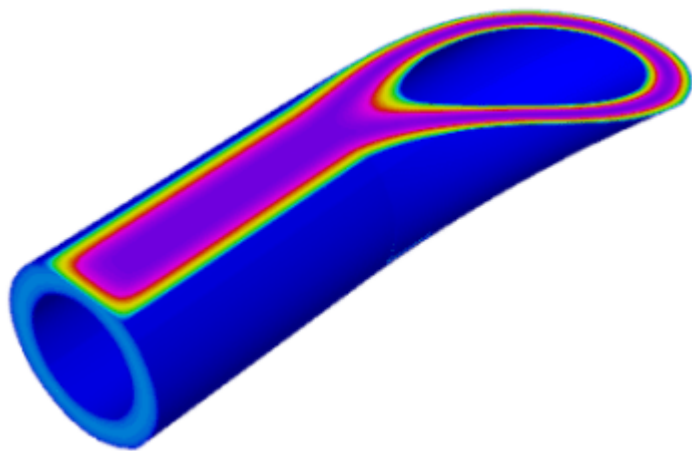


Jan Nikl | NA&S, CASC

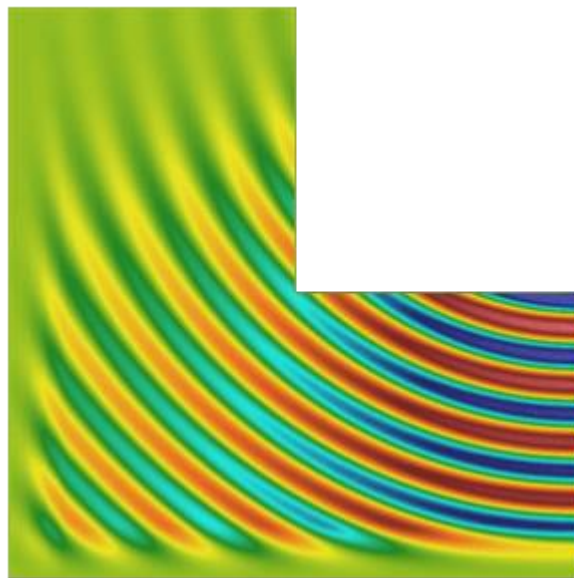
Lawrence Livermore National Laboratory

Mixed systems

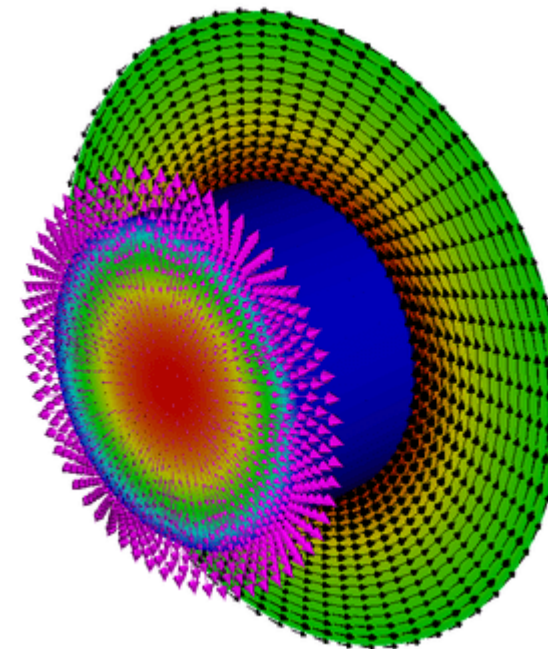
- (In)definite – Darcy, Heat conduction, Maxwell, ...



Example 14
(DG diffusion)



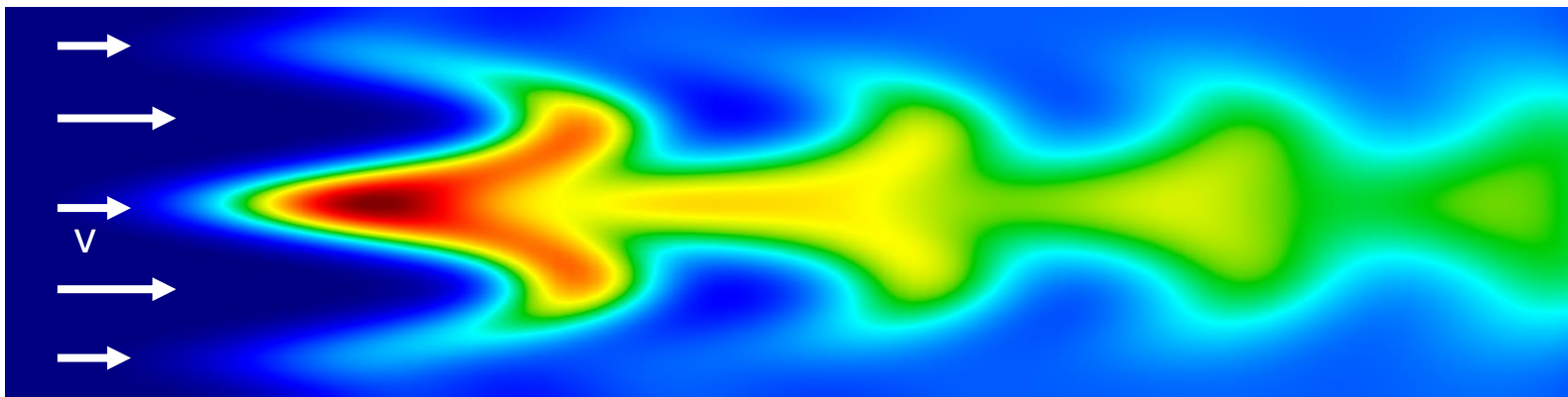
Example 25
(Maxwell problem)



Joule miniapp
(magnetic + thermal diffusion)

Mixed systems

- (In)definite – Darcy, Heat conduction, Maxwell, ...
- Convection-diffusion



Kovaszny flow

Mixed systems

- (In)definite – Darcy, Heat conduction, Maxwell, ...
- Convection-diffusion

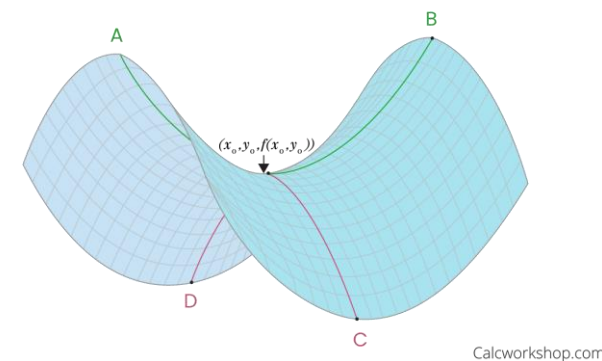
$$\begin{aligned} \underline{\mathbf{q}} + \kappa \nabla u &= 0, \quad \text{in } \Omega, \\ \nabla \cdot (\underline{\mathbf{c}}u + \underline{\mathbf{q}}) &= f, \quad \text{in } \Omega, \end{aligned}$$

- Flux – continuous (RT, ND,...) / discontinuous
- Potential – discontinuous
- Block-(anti)symmetric weak form (not symmetric!)

$$\begin{bmatrix} \underline{A} & -\underline{B}^T \\ \underline{B} & \underline{D} \end{bmatrix}$$

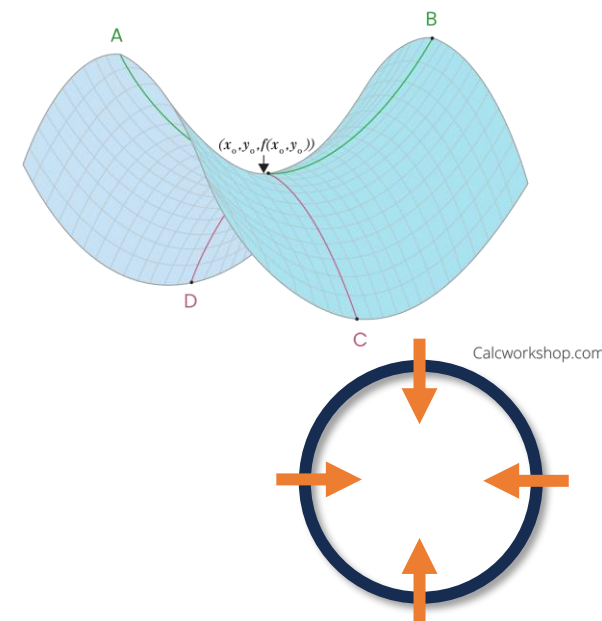
Challenges

- **Definiteness** – saddle-point $\rightarrow$ *primary*?



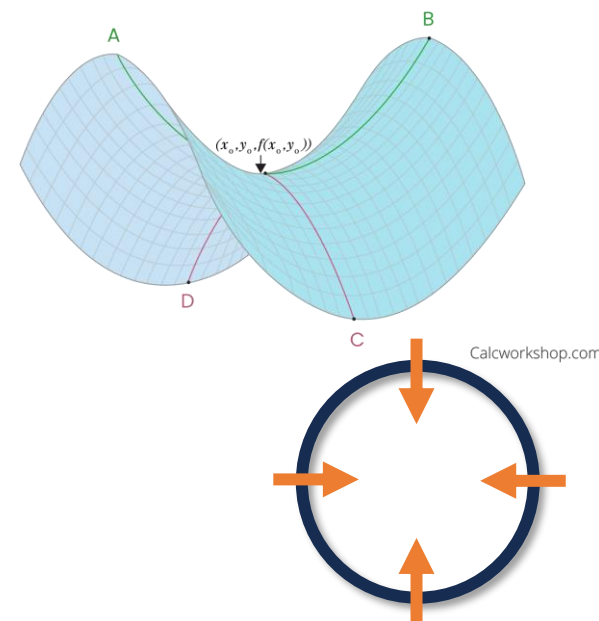
Challenges

- **Definiteness** – saddle-point $\rightarrow$ *primary*?
- **Divergence-free** – potential eq. $\rightarrow$ *mixed*?



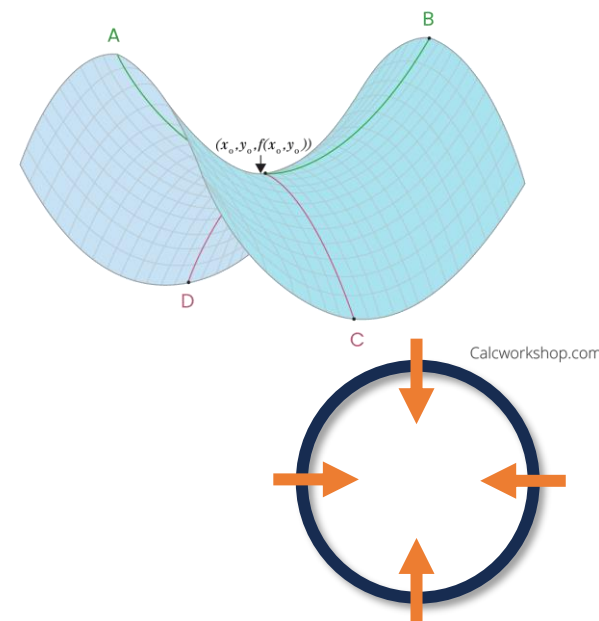
Challenges

- **Definiteness** – saddle-point $\rightarrow$ *primary*?
- **Divergence-free** – potential eq. $\rightarrow$ *mixed*?
- **Conservation** – local cons. of DG potential



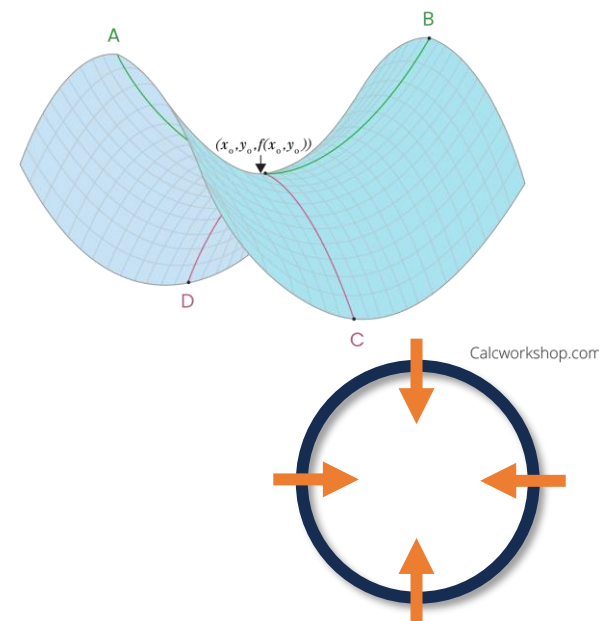
Challenges

- **Definiteness** – saddle-point $\rightarrow$ *primary*?
- **Divergence-free** – potential eq. $\rightarrow$ *mixed*?
- **Conservation** – local cons. of DG potential
- **Preconditioning** – tight coupling $\rightarrow$ *primary*?



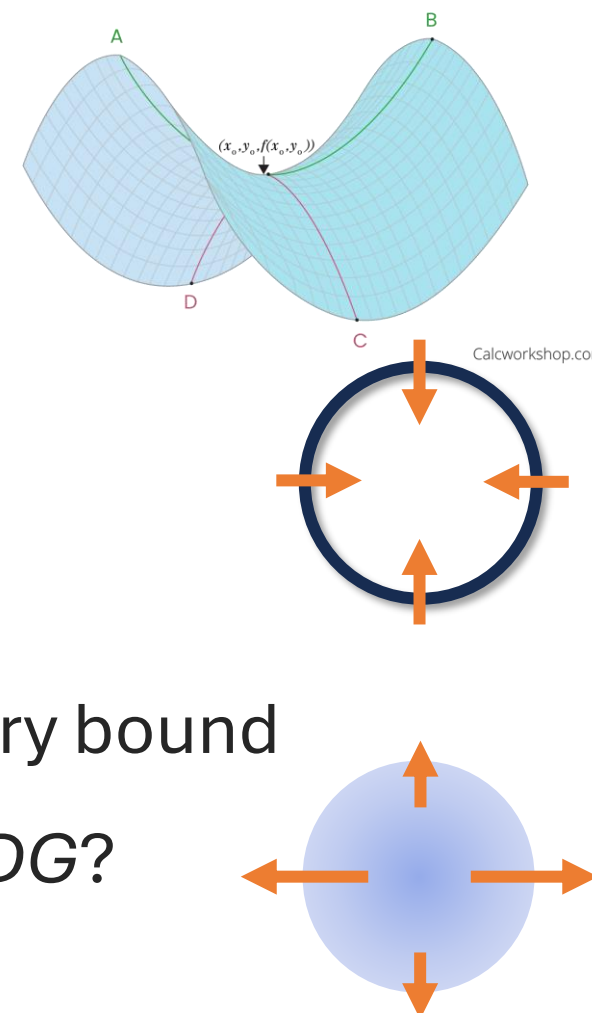
Challenges

- **Definiteness** – saddle-point $\rightarrow$ *primary*?
- **Divergence-free** – potential eq. $\rightarrow$ *mixed*?
- **Conservation** – local cons. of DG potential
- **Preconditioning** – tight coupling $\rightarrow$ *primary*?
- **Memory consumption / data motion** – memory bound



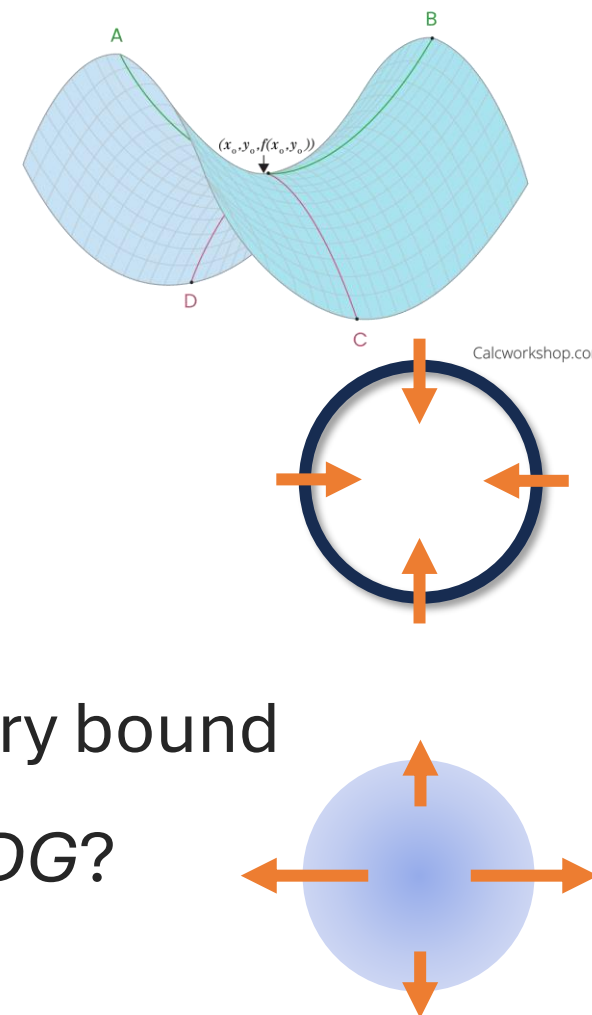
Challenges

- **Definiteness** – saddle-point $\rightarrow$ *primary*?
- **Divergence-free** – potential eq. $\rightarrow$ *mixed*?
- **Conservation** – local cons. of DG potential
- **Preconditioning** – tight coupling $\rightarrow$ *primary*?
- **Memory consumption / data motion** – memory bound
- **Anisotropy** – ringing of CG, preconditioning $\rightarrow$ *DG*?



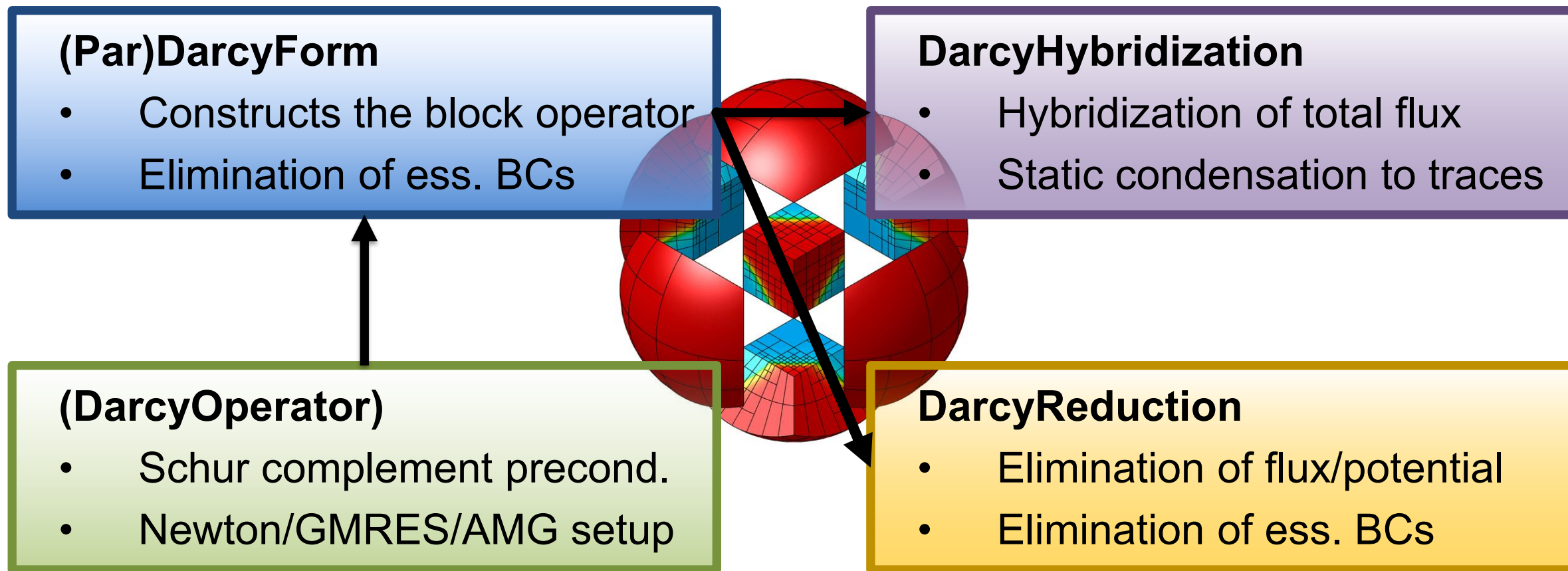
Challenges

- **Definiteness** – saddle-point $\rightarrow$ *primary*?
- **Divergence-free** – potential eq. $\rightarrow$ *mixed*?
- **Conservation** – local cons. of DG potential
- **Preconditioning** – tight coupling $\rightarrow$ *primary*?
- **Memory consumption / data motion** – memory bound
- **Anisotropy** – ringing of CG, preconditioning $\rightarrow$ *DG*?



Best of *primary* and *mixed* formulation? *DGs*? And beyond?

Framework for mixed systems



Mixed formulation – RTDG/LDG

- Finite element method – discrete fxs u_h, q_h test fxs v, w
- (Bi)linear forms (**(Par)DarcyForm**) – $(\bullet, \bullet)$ volume, $\langle \bullet, \bullet \rangle$ face

Mixed formulation – RTDG/LDG

- Finite element method – discrete fxs u_h, q_h test fxs v, w
- (Bi)linear forms (**(Par)DarcyForm**) – $(\bullet, \bullet)$ volume, $\langle \bullet, \bullet \rangle$ face
- *Raviart-Thomas + Discontinuous Galerkin (RTDG)*

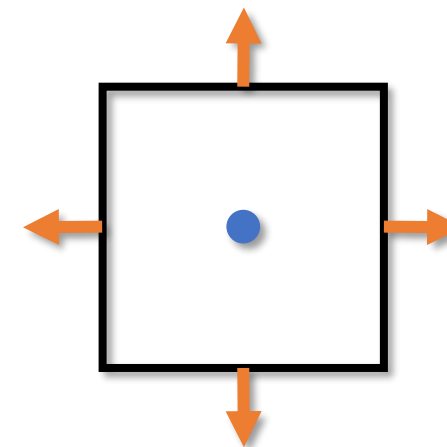
Mixed formulation – RTDG/LDG

- Finite element method – discrete fxs u_h, q_h test fxs v, w
- (Bi)linear forms (**(Par)DarcyForm**) – $(\bullet, \bullet)$ volume, $\langle \bullet, \bullet \rangle$ face
- *Raviart-Thomas + Discontinuous Galerkin (RTDG)*

$$\begin{aligned} & \frac{(\kappa^{-1} q_h, v)_K}{(\nabla \cdot q_h, w)_K} - \frac{(u_h, \nabla \cdot v)_K}{(cu_h, \nabla w)_K} = 0, \\ & \frac{(\nabla \cdot q_h, w)_K}{(cu_h, \nabla w)_K} + \langle \widehat{cu_h} \cdot n, w \rangle_{\partial K} = \frac{(f, w)_K}{(f, w)_K}, \end{aligned}$$

- Schur complement preconditioner (**DarcyOperator**)

$$\begin{bmatrix} A_d^{-1} & \\ & (D + BA_d^{-1}B^T)_{GS}^{-1} \end{bmatrix} \begin{bmatrix} \frac{A}{B} & \frac{-B^T}{D} \end{bmatrix}$$



Example 5 - RTDG (ex5.cpp)

```
DarcyForm *darcy = new DarcyForm(R_space, W_space,
                                   false);

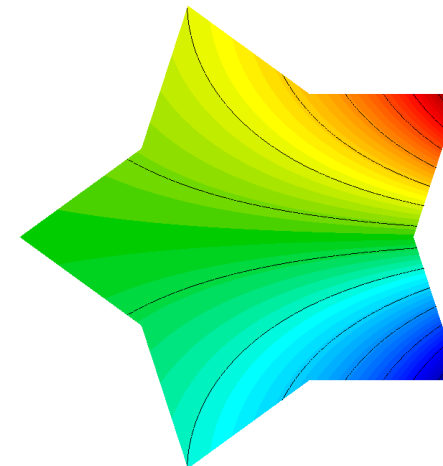
BilinearForm *mVarf = darcy->GetFluxMassForm();
MixedBilinearForm *bVarf = darcy->GetFluxDivForm();

mVarf->AddDomainIntegrator(
    new VectorFEMassIntegrator(kcoeff));

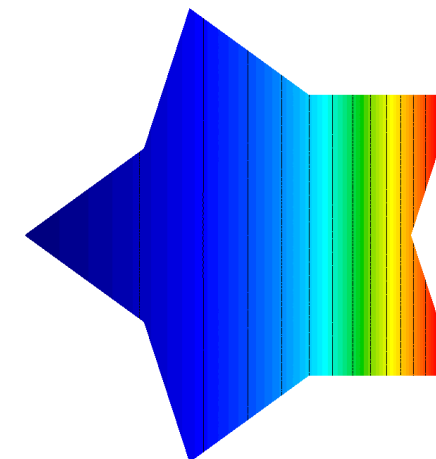
ConstantCoefficient cdiv(-1.);
bVarf->AddDomainIntegrator(
    new VectorFEDivergenceIntegrator(cdiv));

if (pa) { darcy->SetAssemblyLevel(
    AssemblyLevel::PARTIAL); }

darcy->Assemble();
```



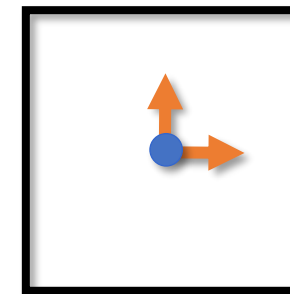
Potential



Flux

Local Discontinuous Galerkin (LDG)

$$\begin{aligned}
 &(\kappa^{-1} \mathbf{q}_h, \mathbf{v})_K - (\mathbf{u}_h, \nabla \cdot \mathbf{v})_K + \langle \hat{\mathbf{u}}_h, \mathbf{v} \cdot \mathbf{n} \rangle_{\partial K} = 0, \quad \forall \mathbf{v} \in (\mathcal{P}^p(K))^d, \\
 &-(\mathbf{c} \mathbf{u}_h + \mathbf{q}_h, \nabla \mathbf{w})_K + \langle (\hat{\mathbf{c}} \mathbf{u}_h + \hat{\mathbf{q}}_h) \cdot \mathbf{n}, \mathbf{w} \rangle_{\partial K} = (\mathbf{f}, \mathbf{w})_K, \quad \forall \mathbf{w} \in \mathcal{P}^p(K).
 \end{aligned}$$

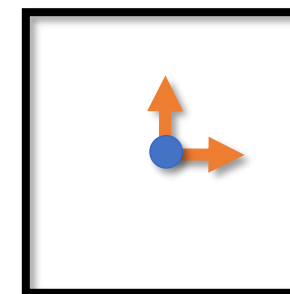


Local Discontinuous Galerkin (LDG)

$$\begin{aligned}
 &(\kappa^{-1} \underline{q}_h, \underline{v})_K - (\underline{u}_h, \nabla \cdot \underline{v})_K + \langle \hat{\underline{u}}_h, \underline{v} \cdot \underline{n} \rangle_{\partial K} = 0, \quad \forall \underline{v} \in (\mathcal{P}^p(K))^d, \\
 & - (\underline{c} \underline{u}_h + \underline{q}_h, \nabla \underline{w})_K + \langle (\hat{\underline{c}} \underline{u}_h + \hat{\underline{q}}_h) \cdot \underline{n}, \underline{w} \rangle_{\partial K} = (\underline{f}, \underline{w})_K, \quad \forall \underline{w} \in \mathcal{P}^p(K).
 \end{aligned}$$

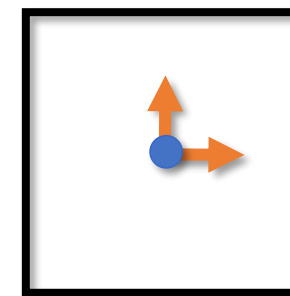
- Traces definition $\rightarrow$ local stabilization

$$\begin{aligned}
 \hat{\underline{q}}_h &= \{\{\underline{q}_h\}\} + C_{11} \llbracket \underline{u}_h \underline{n} \rrbracket + C_{12} \llbracket \underline{q}_h \cdot \underline{n} \rrbracket, \\
 \lambda_h = \hat{\underline{u}}_h &= \{\{\underline{u}_h\}\} - C_{12} \cdot \llbracket \underline{u}_h \underline{n} \rrbracket + C_{22} \llbracket \underline{q}_h \cdot \underline{n} \rrbracket,
 \end{aligned}$$



Local Discontinuous Galerkin (LDG)

$$\begin{aligned}
 &(\kappa^{-1} \underline{q}_h, \underline{v})_K - (\underline{u}_h, \nabla \cdot \underline{v})_K + \langle \hat{\underline{u}}_h, \underline{v} \cdot \underline{n} \rangle_{\partial K} = 0, \quad \forall \underline{v} \in (\mathcal{P}^p(K))^d, \\
 & - (\underline{c} \underline{u}_h + \underline{q}_h, \nabla \underline{w})_K + \langle (\hat{\underline{c}} \underline{u}_h + \hat{\underline{q}}_h) \cdot \underline{n}, \underline{w} \rangle_{\partial K} = (\underline{f}, \underline{w})_K, \quad \forall \underline{w} \in \mathcal{P}^p(K).
 \end{aligned}$$



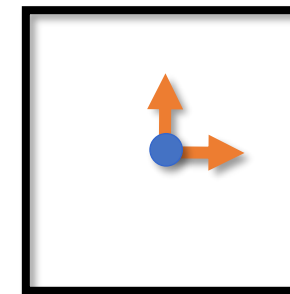
- Traces definition → local stabilization

$$\begin{aligned}
 \hat{\underline{q}}_h &= \{\{\underline{q}_h\}\} + C_{11} \llbracket \underline{u}_h \underline{n} \rrbracket + C_{12} \llbracket \underline{q}_h \cdot \underline{n} \rrbracket, \\
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 \end{aligned}$$

- LDG: $C_{22}=0$ (flux elimination → **DarcyReduction** $(D + BA^{-1}B^T)u_h = f$)

Local Discontinuous Galerkin (LDG)

$$\begin{aligned}
 &(\kappa^{-1} q_h, v)_K - (u_h, \nabla \cdot v)_K + \langle \hat{u}_h, v \cdot n \rangle_{\partial K} = 0, \quad \forall v \in (\mathcal{P}^p(K))^d, \\
 &-(cu_h + q_h, \nabla w)_K + \langle (\hat{cu}_h + \hat{q}_h) \cdot n, w \rangle_{\partial K} = (f, w)_K, \quad \forall w \in \mathcal{P}^p(K).
 \end{aligned}$$



- Traces definition → local stabilization

$$\begin{aligned}
 \hat{q}_h &= \{\{q_h\}\} + C_{11} \llbracket u_h n \rrbracket + C_{12} \llbracket q_h \cdot n \rrbracket, \\
 \lambda_h = \hat{u}_h &= \{\{u_h\}\} - C_{12} \llbracket u_h n \rrbracket + C_{22} \llbracket q_h \cdot n \rrbracket,
 \end{aligned}$$

- LDG: $C_{22}=0$ (flux elimination → **DarcyReduction** $(D + BA^{-1}B^T)u_h = f$)
- Centered scheme: $C_{12}=0, C_{11}=\kappa h^{-1}/2$

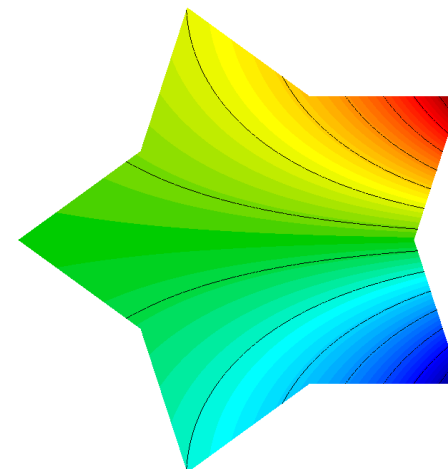
$$\begin{aligned}
 &(\kappa^{-1} q_h, v) - (u_h, \nabla \cdot v) + \langle \{\{u_h\}\}, \llbracket v \cdot n \rrbracket \rangle = 0, \\
 &(\nabla \cdot q_h, w) - \langle \llbracket q_h \cdot n \rrbracket, \{\{w\}\} \rangle + \underbrace{\langle \frac{\kappa h^{-1}}{2} \llbracket u_h \rrbracket, \llbracket v \rrbracket \rangle}_{\approx \text{DG diffusion}} = (f, w)
 \end{aligned}$$

Example 5 – LDG (hdg/ex5.cpp)

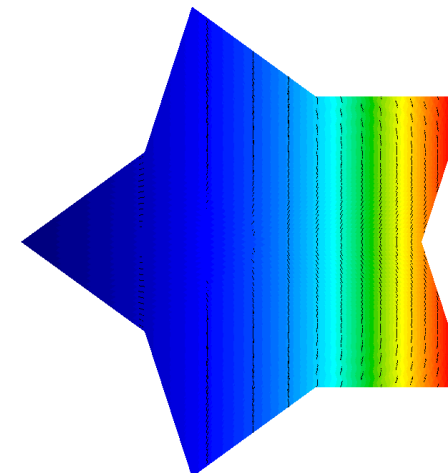
```
DarcyForm *darcy = new DarcyForm(R_space, W_space);
BilinearForm *mVarf = darcy->GetFluxMassForm();
MixedBilinearForm *bVarf = darcy->GetFluxDivForm();
BilinearForm *mtVarf = GetPotentialMassForm();

mVarf->AddDomainIntegrator(new
    VectorMassIntegrator(kcoeff));
bVarf->AddDomainIntegrator(new
    VectorDivergenceIntegrator());
bVarf->AddInteriorFaceIntegrator(new
    TransposeIntegrator(new
        DGNormalTraceIntegrator(-1.)));
mtVarf->AddInteriorFaceIntegrator(new
    HDGDiffusionIntegrator(ikcoeff));

darcy->EnableFluxReduction();
darcy->Assemble();
```

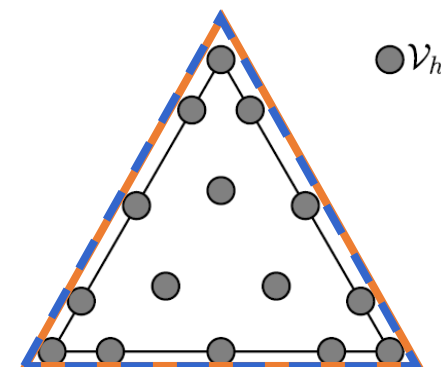


Potential



Flux

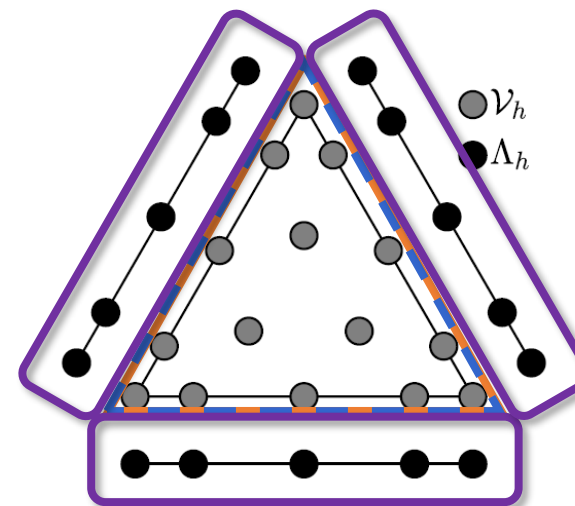
Hybridization



Credit: G. Giorgiani et al. / CPC 254 (2020) 107375

Hybridization

- Lagrange multipliers $\lambda_h \approx \hat{u}_h$



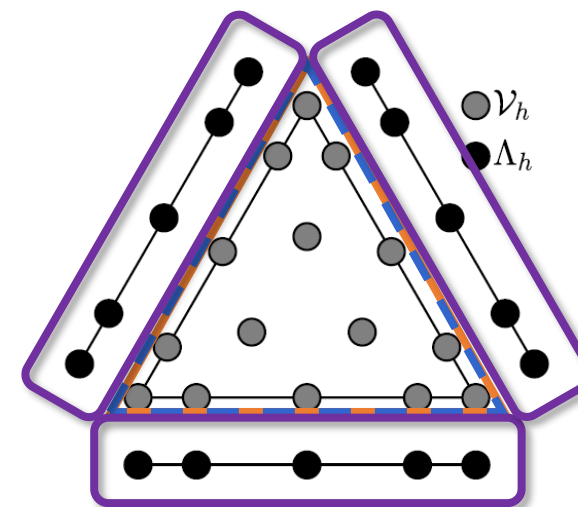
Credit: G. Giorgiani et al. / CPC 254 (2020) 107375

Hybridization

- Lagrange multipliers $\lambda_h \approx \hat{u}_h$
- Weak continuity of total flux

$$\langle \llbracket (\widehat{cu}_h + \widehat{q}_h) \cdot n \rrbracket, \mu \rangle_{\varepsilon_h} = 0, \quad \forall \mu$$

- *N.C. Nguyen, J. Peraire & B. Cockburn (2009), JCP, 228, 3232–3254.*

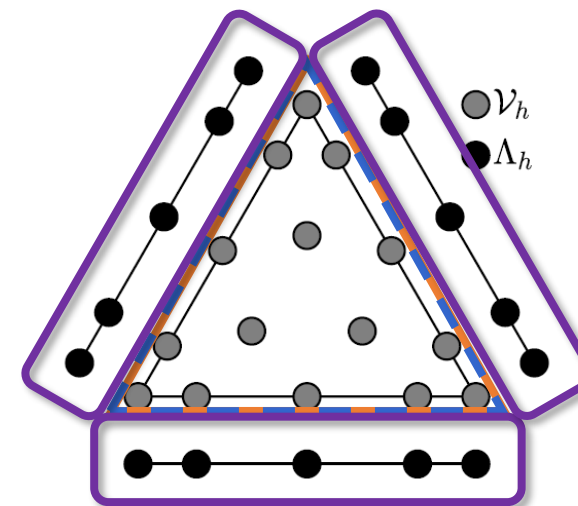


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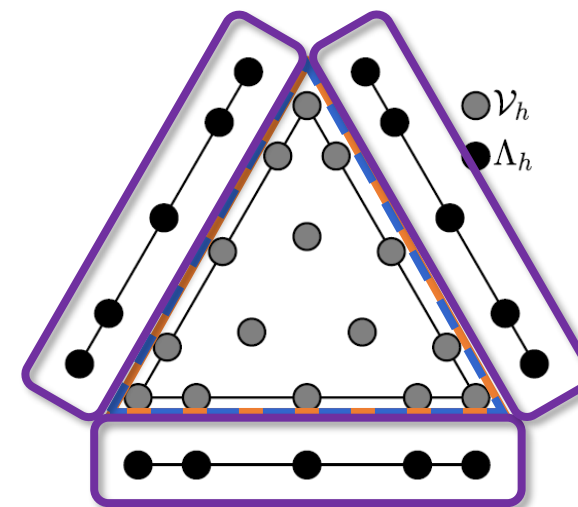
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- Reduction to trace DOFs of λ_h (**DarcyHybridization**)

Hybridization

- Lagrange multipliers $\lambda_h \approx \hat{u}_h$
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- *N.C. Nguyen, J. Peraire & B. Cockburn (2009), JCP, 228, 3232–3254.*

Credit: G. Giorgiani et al. / CPC 254 (2020) 107375

- Reduction to trace DOFs of λ_h (**DarcyHybridization**)
- darcy-hdg-dev
- [PR #4350](#)

[WIP] Hybridization of mixed systems (HRT, HDG)
[darcy-hdg-dev] #4350

Open najlkin wants to merge 497 commits into master from darcy-hdg-dev

Hybridized Raviart-Thomas (HRT)

- Lagrange multiplier $\lambda_h \approx \hat{u}_h$ (`EnableHybridization()`)

$$(\kappa^{-1} \mathbf{q}_h, \mathbf{v})_{T_h} - (\mathbf{u}_h, \nabla \cdot \mathbf{v})_{T_h} + \langle \lambda_h, \mathbf{v} \cdot \mathbf{n} \rangle_{\partial T_h} = 0, \quad \forall \mathbf{v} \in \mathbf{V}_h^p,$$

$$(\nabla \cdot \mathbf{q}_h, w)_{T_h} = (f, w)_{T_h}, \quad \forall w \in W_h^p,$$

$$\langle [\![\hat{\mathbf{q}}_h \cdot \mathbf{n}]\!], \mu \rangle_{\mathcal{E}_h} = 0, \quad \forall \mu \in M_h^p(0).$$

Hybridized Raviart-Thomas (HRT)

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$$(\kappa^{-1} \mathbf{q}_h, \mathbf{v})_{T_h} - (\mathbf{u}_h, \nabla \cdot \mathbf{v})_{T_h} + \langle \lambda_h, \mathbf{v} \cdot \mathbf{n} \rangle_{\partial T_h} = 0, \quad \forall \mathbf{v} \in \mathbf{V}_h^p,$$

$$(\nabla \cdot \mathbf{q}_h, \mathbf{w})_{T_h} = (\mathbf{f}, \mathbf{w})_{T_h}, \quad \forall \mathbf{w} \in \mathbf{W}_h^p,$$

$$\langle [\![\hat{\mathbf{q}}_h \cdot \mathbf{n}]\!], \mu \rangle_{\mathcal{E}_h} = 0, \quad \forall \mu \in M_h^p(0).$$

- Reduction of the system: (`FormLinearSystem()`)

$$\begin{bmatrix} \underline{A} & \underline{-B^T} & \underline{C^T} \\ \underline{B} & 0 & 0 \\ \underline{C} & 0 & 0 \end{bmatrix} \begin{bmatrix} \underline{Q} \\ \underline{U} \\ \underline{\Lambda} \end{bmatrix} = \begin{bmatrix} 0 \\ \underline{F} \\ 0 \end{bmatrix}. \quad \Rightarrow \quad \begin{aligned} \mathbb{K} &= -[C \ 0] \begin{bmatrix} A & -B^T \\ B & 0 \end{bmatrix}^{-1} \begin{bmatrix} C^T \\ 0 \end{bmatrix} \\ \mathbb{F} &= -[C \ 0] \begin{bmatrix} A & -B^T \\ B & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ F \end{bmatrix} \end{aligned} \quad \Rightarrow \quad \boxed{\mathbb{K} \Lambda = \mathbb{F}},$$

Hybridized Raviart-Thomas (HRT)

- Lagrange multiplier $\lambda_h \approx \hat{u}_h$ (`EnableHybridization()`)

$$(\kappa^{-1} \mathbf{q}_h, \mathbf{v})_{T_h} - (\mathbf{u}_h, \nabla \cdot \mathbf{v})_{T_h} + \langle \lambda_h, \mathbf{v} \cdot \mathbf{n} \rangle_{\partial T_h} = 0, \quad \forall \mathbf{v} \in \mathbf{V}_h^p,$$

$$(\nabla \cdot \mathbf{q}_h, \mathbf{w})_{T_h} = (\mathbf{f}, \mathbf{w})_{T_h}, \quad \forall \mathbf{w} \in \mathbf{W}_h^p,$$

$$\langle [\![\hat{\mathbf{q}}_h \cdot \mathbf{n}]\!], \mu \rangle_{\mathcal{E}_h} = 0, \quad \forall \mu \in M_h^p(0).$$

- Reduction of the system: (`FormLinearSystem()`)

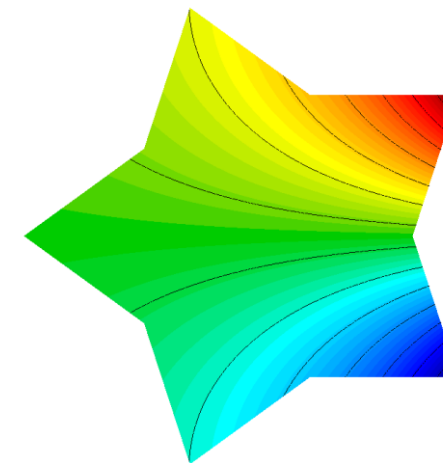
$$\begin{bmatrix} \underline{A} & \underline{-B^T} & \underline{C^T} \\ \underline{B} & 0 & 0 \\ \underline{C} & 0 & 0 \end{bmatrix} \begin{bmatrix} \underline{Q} \\ \underline{U} \\ \underline{\Lambda} \end{bmatrix} = \begin{bmatrix} 0 \\ \underline{F} \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} \mathbb{K} = -[C \ 0] \begin{bmatrix} A & -B^T \\ B & 0 \end{bmatrix}^{-1} \begin{bmatrix} C^T \\ 0 \end{bmatrix} \\ \mathbb{F} = -[C \ 0] \begin{bmatrix} A & -B^T \\ B & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ F \end{bmatrix} \end{matrix}, \Rightarrow \boxed{\mathbb{K} \Lambda = \mathbb{F}},$$

- Recovery of the solution: (`RecoverFEMSolution()`)

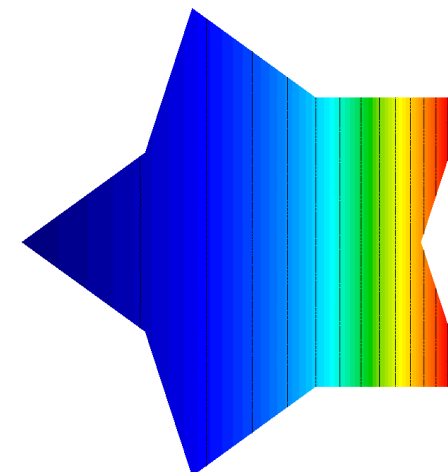
$$\begin{bmatrix} \underline{Q} \\ \underline{U} \end{bmatrix} = \begin{bmatrix} A & -B^T \\ B & 0 \end{bmatrix}^{-1} \left(\begin{bmatrix} 0 \\ F \end{bmatrix} - \begin{bmatrix} C^T \\ 0 \end{bmatrix} \Lambda \right),$$

Example 5 – HRT (ex5.cpp)

```
...
if (hybridization)
{
    trace_coll = new DG_Interface_FECollection(
        order, dim);
    trace_space = new FiniteElementSpace(
        mesh, trace_coll);
    darcy->EnableHybridization(trace_space,
        new NormalTraceJumpIntegrator(),
        ess_flux_tdofs_list);
}
darcy->Assemble();
```



Potential



Flux

Hybridizable Discontinuous Galerkin (HDG)

- Lagrange multiplier $\lambda_h \approx \hat{u}_h$

$$(\kappa^{-1} \mathbf{q}_h, \mathbf{v})_{T_h} - (\mathbf{u}_h, \nabla \cdot \mathbf{v})_{T_h} + \langle \lambda_h, \mathbf{v} \cdot \mathbf{n} \rangle_{\partial T_h} = 0, \quad \forall \mathbf{v} \in V_h^p,$$

$$-(c \mathbf{u}_h + \mathbf{q}_h, \nabla \mathbf{w})_{T_h} + \langle (\hat{c} \mathbf{u}_h + \hat{\mathbf{q}}_h) \cdot \mathbf{n}, \mathbf{w} \rangle_{\partial T_h} = (f, \mathbf{w})_{T_h}, \quad \forall \mathbf{w} \in W_h^p,$$

$$\langle [(\hat{c} \mathbf{u}_h + \hat{\mathbf{q}}_h) \cdot \mathbf{n}], \mu \rangle_{\mathcal{E}_h} = 0, \quad \forall \mu \in M_h^p(0).$$

- *N.C. Nguyen, J. Peraire & B. Cockburn (2009), JCP, 228, 3232–3254.*

Hybridizable Discontinuous Galerkin (HDG)

- Lagrange multiplier $\lambda_h \approx \hat{u}_h$

$$(\kappa^{-1} \mathbf{q}_h, \mathbf{v})_{T_h} - (\mathbf{u}_h, \nabla \cdot \mathbf{v})_{T_h} + \langle \lambda_h, \mathbf{v} \cdot \mathbf{n} \rangle_{\partial T_h} = 0, \quad \forall \mathbf{v} \in V_h^p,$$

$$-(\mathbf{c} \mathbf{u}_h + \mathbf{q}_h, \nabla \mathbf{w})_{T_h} + \langle (\hat{\mathbf{c}} \mathbf{u}_h + \hat{\mathbf{q}}_h) \cdot \mathbf{n}, \mathbf{w} \rangle_{\partial T_h} = (f, \mathbf{w})_{T_h}, \quad \forall \mathbf{w} \in W_h^p,$$

$$\langle \llbracket (\hat{\mathbf{c}} \mathbf{u}_h + \hat{\mathbf{q}}_h) \cdot \mathbf{n} \rrbracket, \mu \rangle_{\mathcal{E}_h} = 0, \quad \forall \mu \in M_h^p(0).$$

- *N.C. Nguyen, J. Peraire & B. Cockburn (2009), JCP, 228, 3232–3254.*
- Stabilization: $\hat{\mathbf{c}} \mathbf{u}_h + \hat{\mathbf{q}}_h = \mathbf{c} \mathbf{u}_h + \mathbf{q}_h + \tau(\mathbf{u}_h - \hat{\mathbf{u}}_h) \mathbf{n}, \rightarrow \text{Centered/upwind}$

Hybridizable Discontinuous Galerkin (HDG)

- Lagrange multiplier $\lambda_h \approx \hat{u}_h$

$$(\kappa^{-1} \mathbf{q}_h, \mathbf{v})_{T_h} - (\mathbf{u}_h, \nabla \cdot \mathbf{v})_{T_h} + \langle \lambda_h, \mathbf{v} \cdot \mathbf{n} \rangle_{\partial T_h} = 0, \quad \forall \mathbf{v} \in \mathbf{V}_h^p,$$

$$-(\mathbf{c} \mathbf{u}_h + \mathbf{q}_h, \nabla \mathbf{w})_{T_h} + \langle (\hat{\mathbf{c}} \mathbf{u}_h + \hat{\mathbf{q}}_h) \cdot \mathbf{n}, \mathbf{w} \rangle_{\partial T_h} = (f, \mathbf{w})_{T_h}, \quad \forall \mathbf{w} \in \mathbf{W}_h^p,$$

$$\langle [(\hat{\mathbf{c}} \mathbf{u}_h + \hat{\mathbf{q}}_h) \cdot \mathbf{n}], \mu \rangle_{\mathcal{E}_h} = 0, \quad \forall \mu \in M_h^p(0).$$

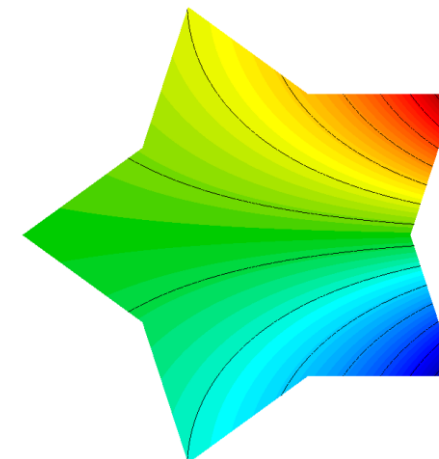
- N.C. Nguyen, J. Peraire & B. Cockburn (2009), JCP, 228, 3232–3254.*
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- Reduction of the system:

$$\begin{bmatrix} \mathbf{A} & -\mathbf{B}^T & \mathbf{C}^T \\ \mathbf{B} & \mathbf{D} & \mathbf{E} \\ \mathbf{C} & \mathbf{G} & \mathbf{H} \end{bmatrix} \begin{bmatrix} \mathbf{Q} \\ \mathbf{U} \\ \mathbf{A} \end{bmatrix} = \begin{bmatrix} \mathbf{R} \\ \mathbf{F} \\ \mathbf{L} \end{bmatrix} \Rightarrow \mathbb{K} = -[\mathbf{C} \quad \mathbf{G}] \begin{bmatrix} \mathbf{A} & -\mathbf{B}^T \\ \mathbf{B} & \mathbf{D} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{C}^T \\ \mathbf{E} \end{bmatrix} + \mathbf{H},$$

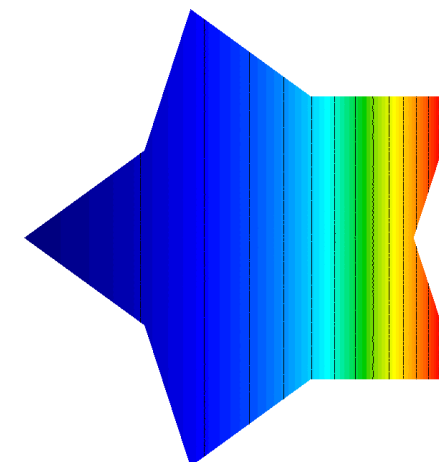
$$\mathbb{F} = \mathbf{L} - [\mathbf{C} \quad \mathbf{G}] \begin{bmatrix} \mathbf{A} & -\mathbf{B}^T \\ \mathbf{B} & \mathbf{D} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{R} \\ \mathbf{F} \end{bmatrix} \Rightarrow \boxed{\mathbb{K} \mathbf{A} = \mathbb{F},}$$

Example 5 – HDG (hdg/ex5.cpp)

```
...
BilinearForm *mtVarf = GetPotentialMassForm();
...
mtVarf->AddInteriorFaceIntegrator(new
    HDGDiffusionIntegrator(ikcoeff));
if (hybridization)
{
    trace_coll = new DG_Interface_FECollection(
        order, dim);
    trace_space = new FiniteElementSpace(
        mesh, trace_coll);
    darcy->EnableHybridization(trace_space,
        new NormalTraceJumpIntegrator(),
        ess_flux_tdofs_list);
}
darcy->Assemble();
```



Potential



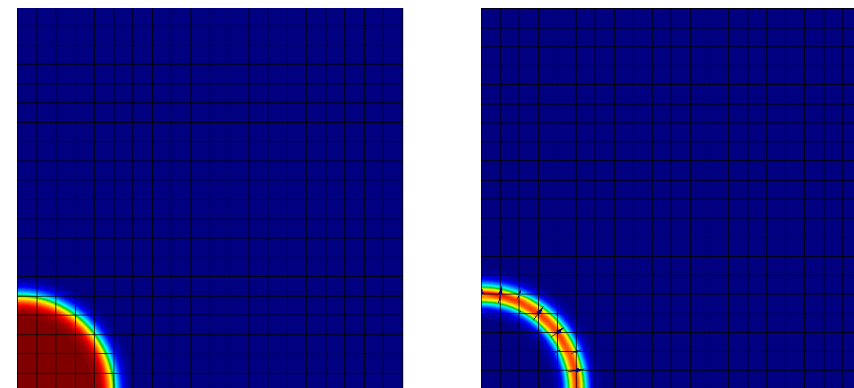
Flux

Convection-diffusion miniapp (convdiff.cpp)

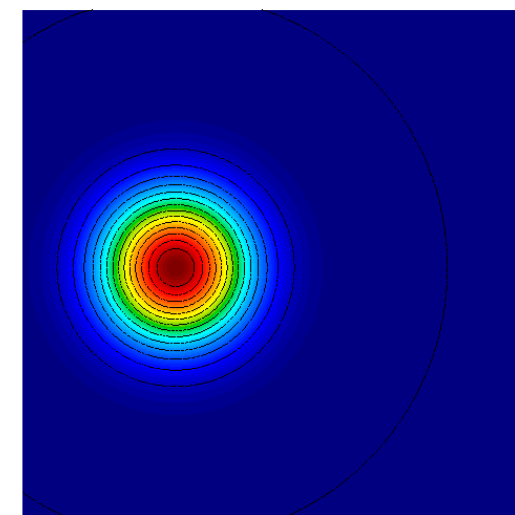
```
...
BilinearForm *Mt = GetPotentialMassForm();
...

Mt->AddDomainIntegrator(
    new ConservativeConvectionIntegrator(ccoeff));

if (upwinded) {
    Mt->AddInteriorFaceIntegrator(
        new HDGConvectionUpwindedIntegrator(ccoeff))
} else {
    Mt->AddInteriorFaceIntegrator(
        new HDGConvectionCenteredIntegrator(ccoeff))
}
}
```



Problem 3 – steady advection

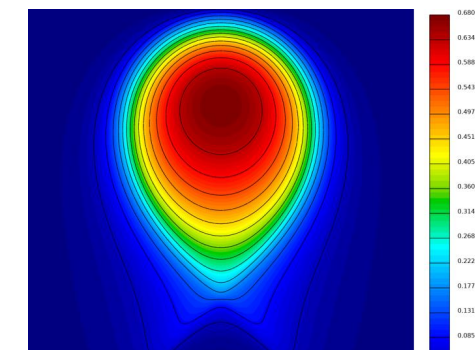
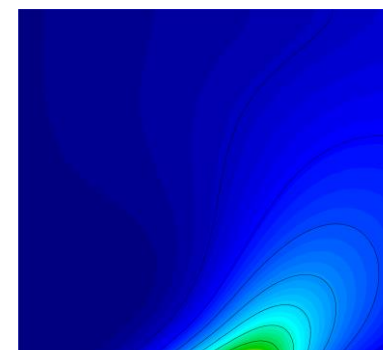


Problem 4 – non-steady advection(-diffusion)

Miniapps & examples

• Plasma

- `braginskii_hdg`
- `scatter`
- `umansky_hdg`
- `stix` (1D/2D) (Mark Stowell)
- `umansky` (Mark Stowell)
- `electrostatic-pic` (Rushan Zhang)



• HDG

- `(p)convdiff` ((non)-steady adv-dif, Kovasznay, Burgers, nonlin. conductivity)
- `(p)anisodiff` (steady sine, diff. ring arc/Gauss/sine, boundary layer, steady peak, Sovinec, Umansky, single/double null)
- `mfem-logo`



Miniapps & examples

• Plasma

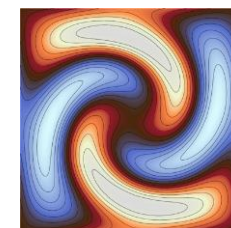
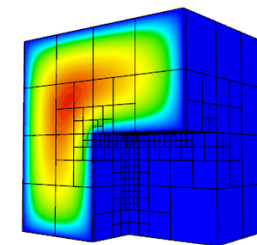
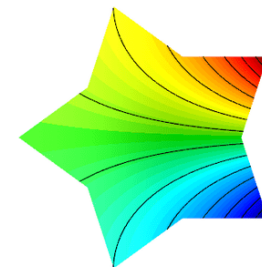
- `braginskii_hdg`
- `scatter`
- `umansky_hdg`
- `stix` (1D/2D) (Mark Stowell)
- `umansky` (Mark Stowell)
- `electrostatic-pic` (Rushan Zhang)

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- `mfem-logo`

• HDG Examples

- `ex5` – Darcy problem
- `ex6` – Poisson AMR
- `ex14` – (H)DG Poisson
- `ex17` – (H)DG elasticity
- `ex21` – elasticity AMR
- `ex41` – IMEX adv-dif



Miniapps & examples

• Plasma

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- `umansky_hdg`
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• HDG

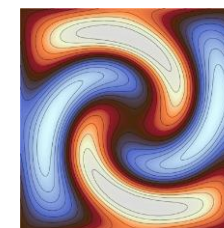
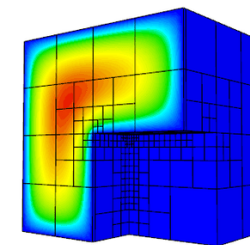
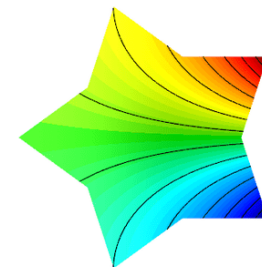
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- `ex6` – Poisson AMR
- `ex14` – (H)DG Poisson
- `ex17` – (H)DG elasticity
- `ex21` – elasticity AMR
- `ex41` – IMEX adv-dif

• Regression tests

- 129 serial
- 96 parallel

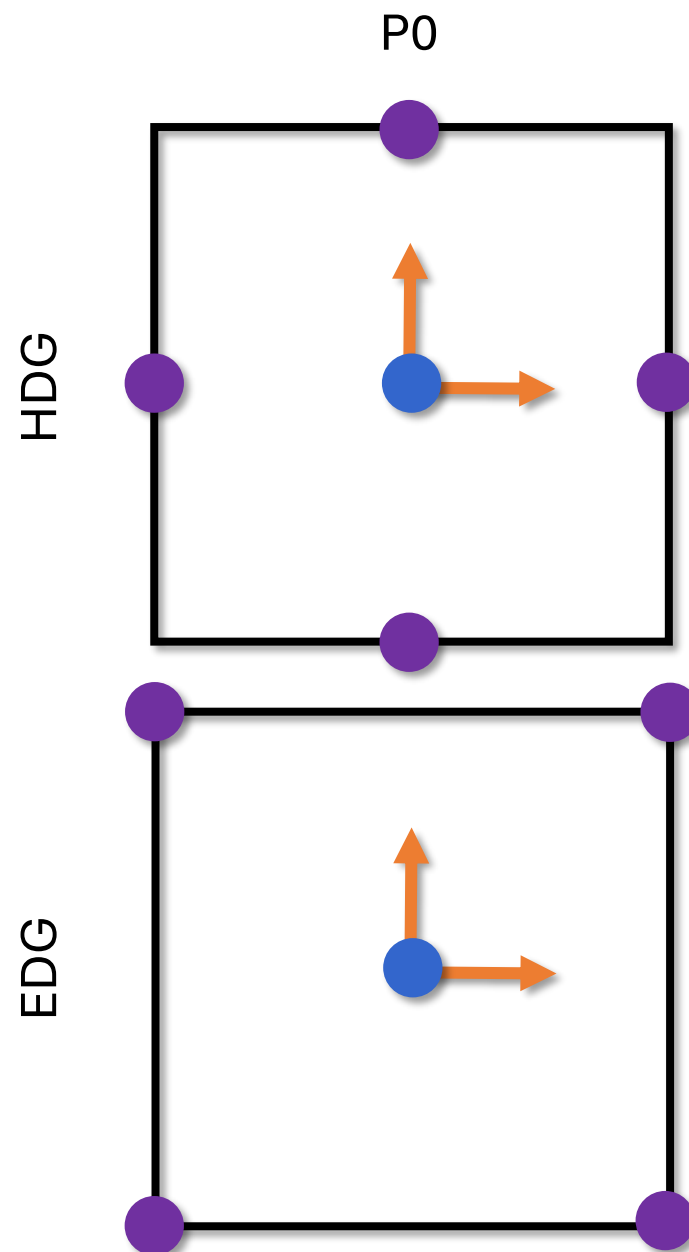




Updates

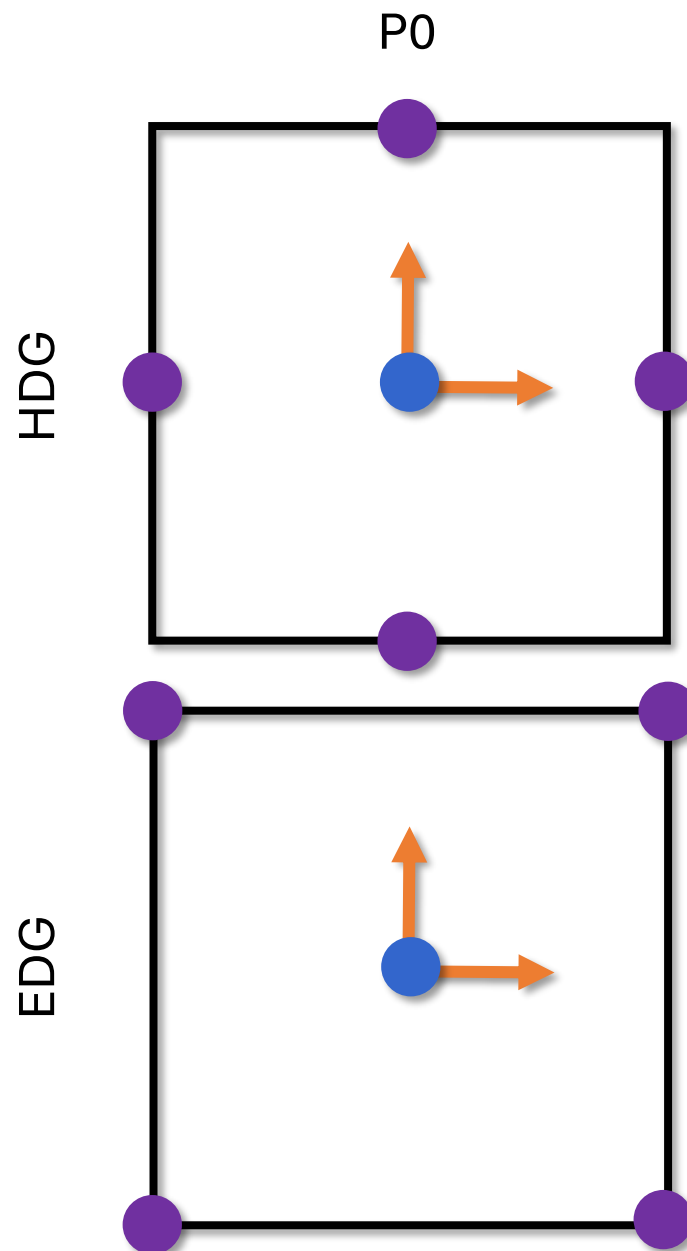
Embedded DG

- DG trace vs. H^1 trace



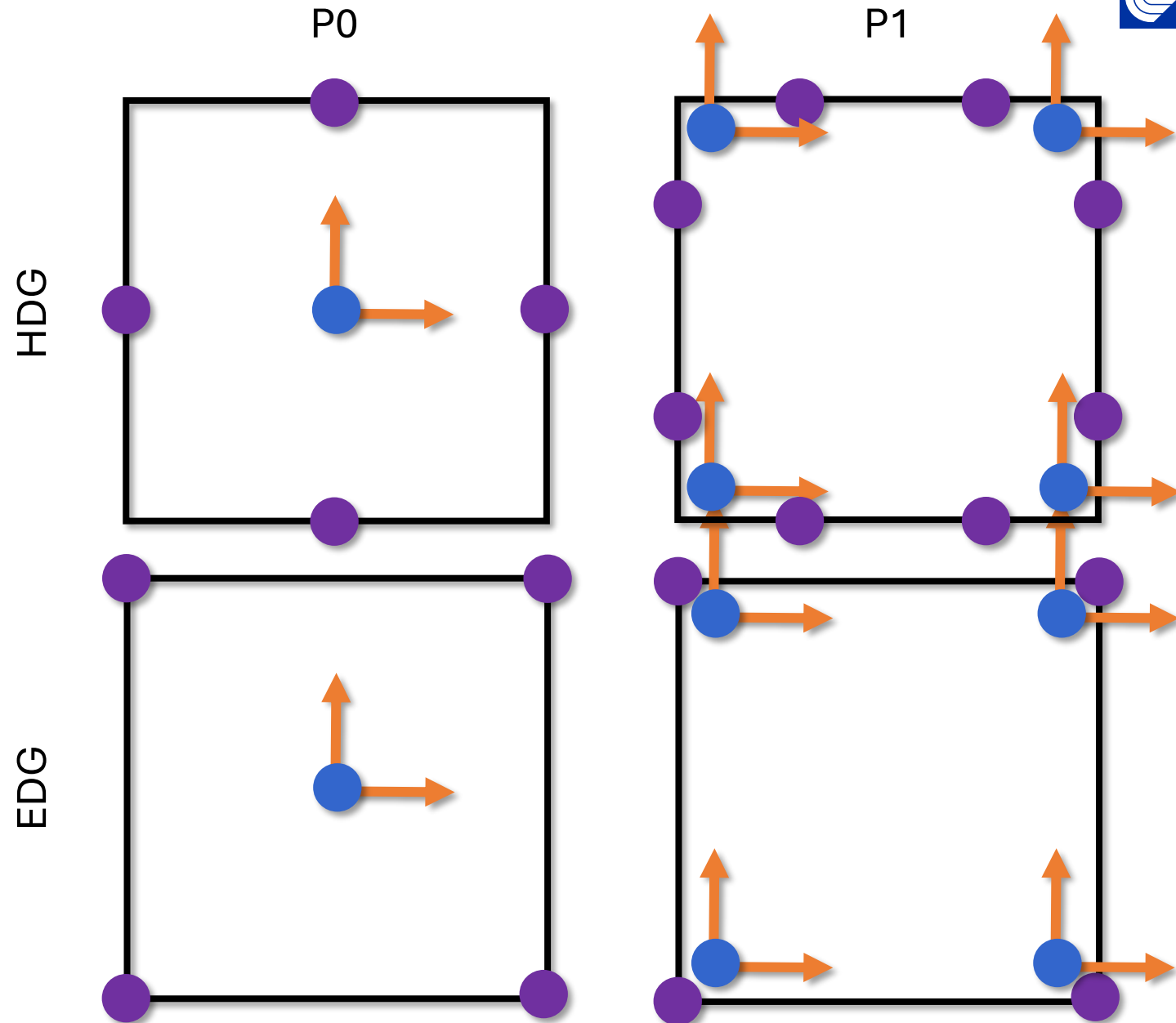
Embedded DG

- DG trace vs. H^1 trace
- Gauss-Lobatto
- Less DOFs



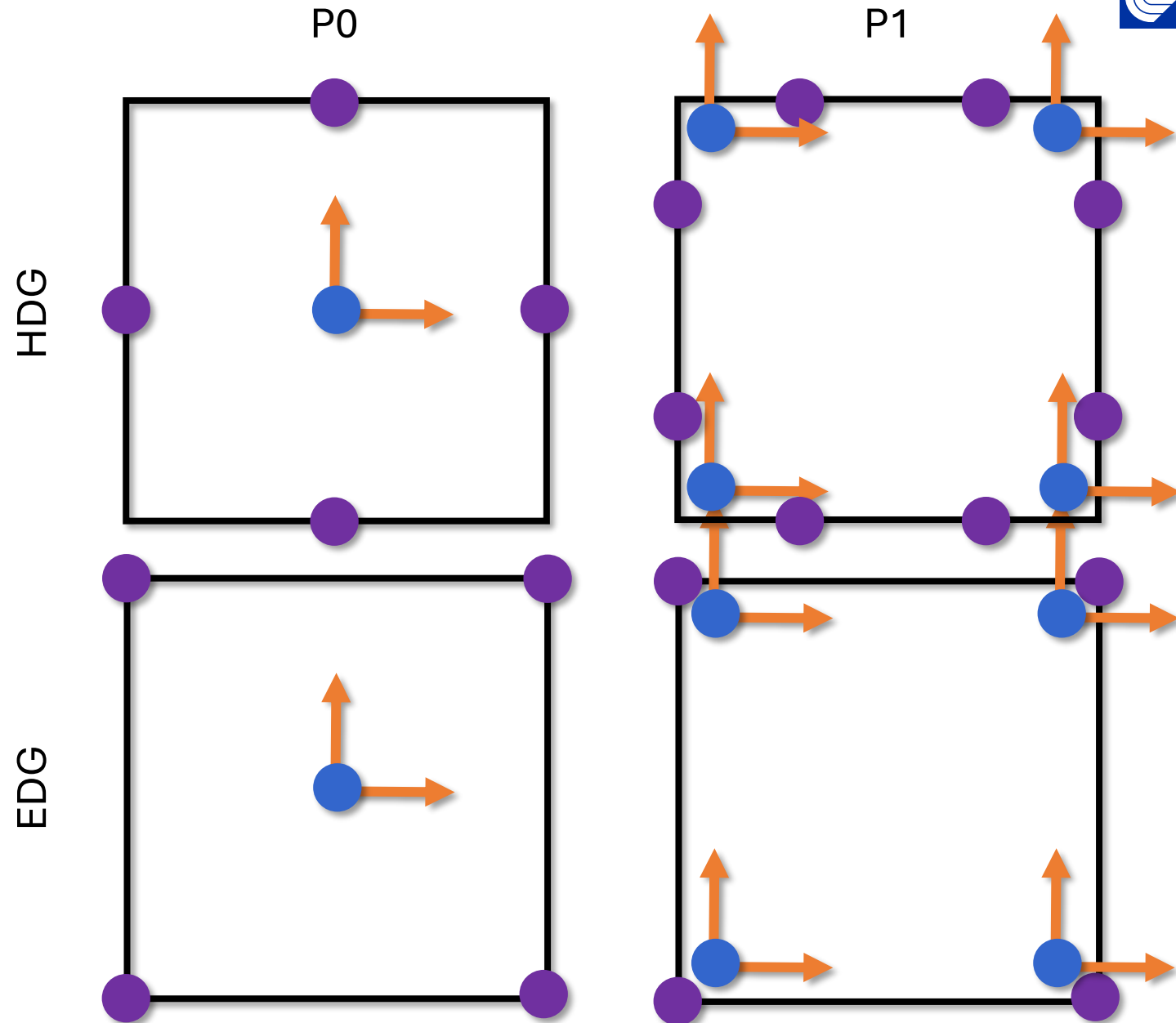
Embedded DG

- DG trace vs. H^1 trace
- Gauss-Lobatto
- Less DOFs



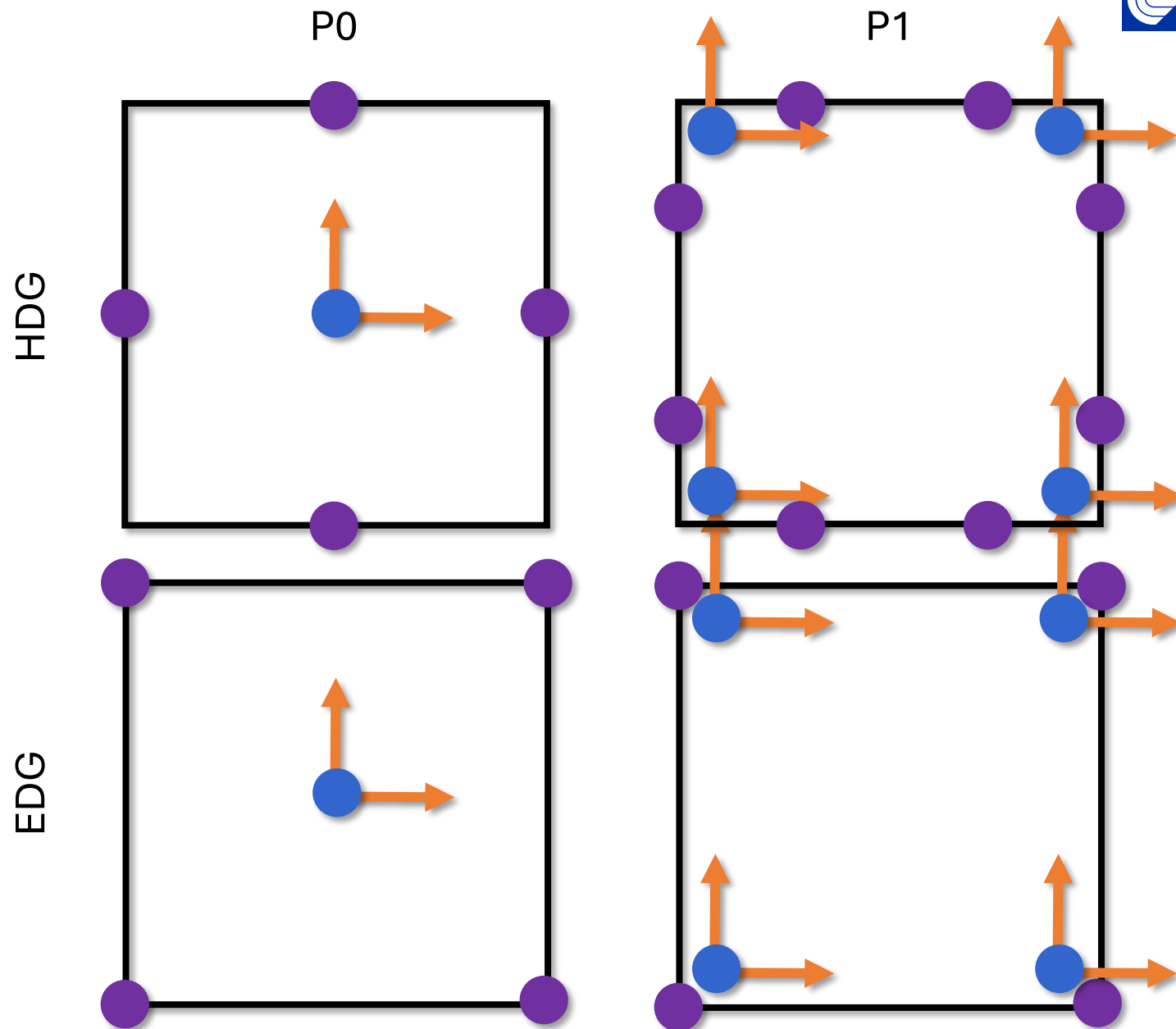
Embedded DG

- DG trace vs. H^1 trace
- Gauss-Lobatto
- Less DOFs
- Conditioning?



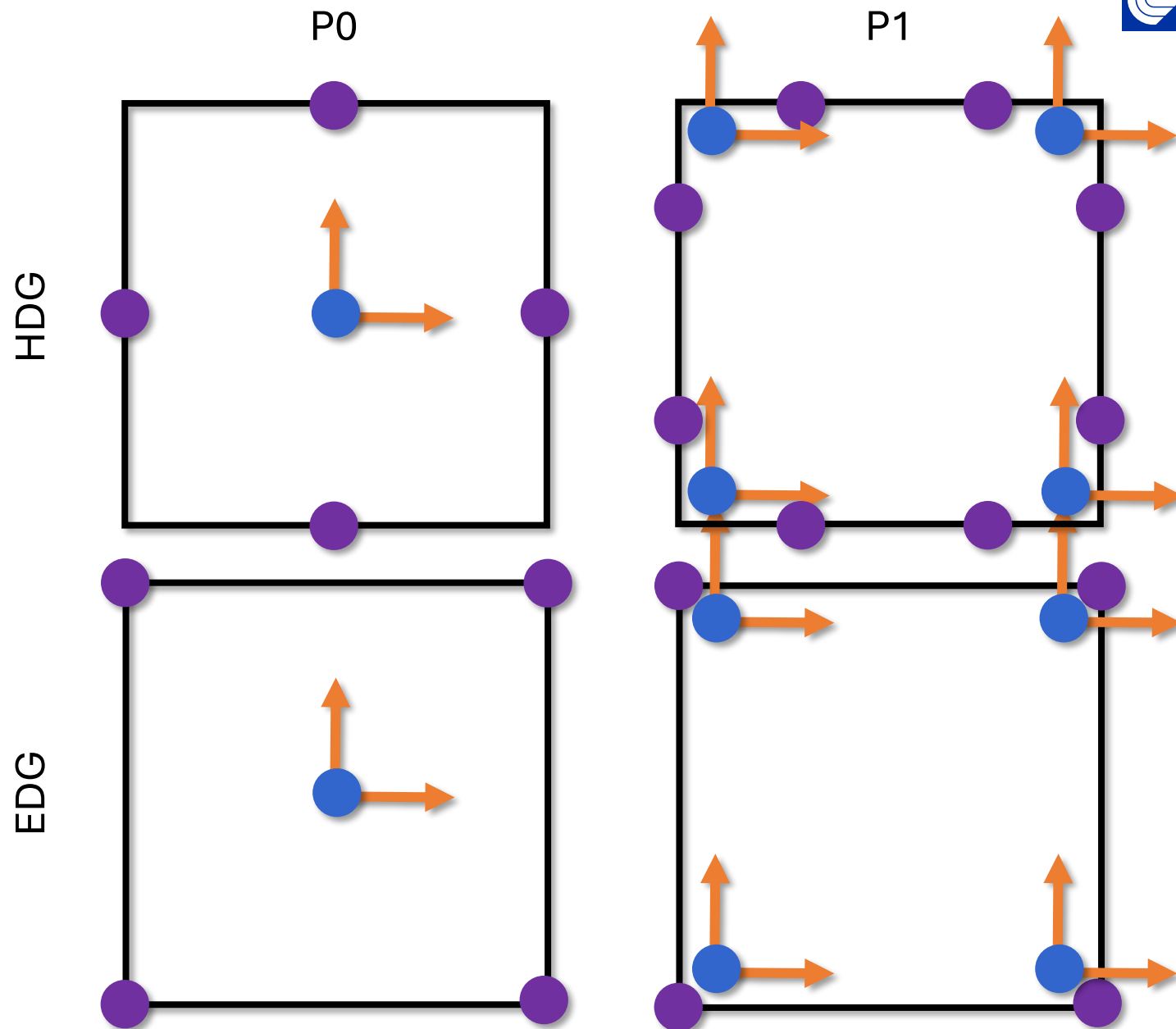
Embedded DG

- DG trace vs. H^1 trace
- Gauss-Lobatto
- Less DOFs
- Conditioning?
- Compatibility with AMG



Embedded DG

- DG trace vs. H^1 trace
- Gauss-Lobatto
- Less DOFs
- Conditioning?
- Compatibility with AMG
- No reconstruction

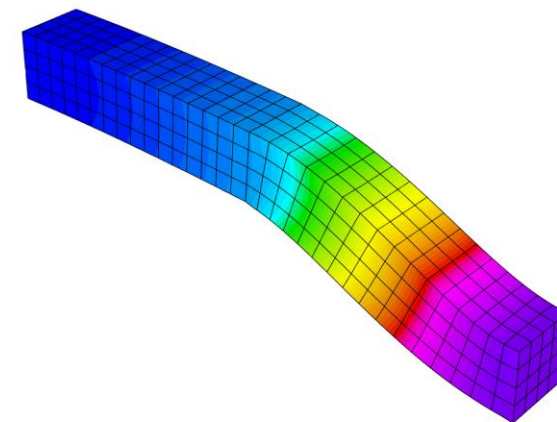


Elasticity

- Stress σ , displacement u , Lamé coefficients λ, μ

$$\underline{\underline{\sigma}} = \lambda \nabla \cdot \underline{u} I + \mu (\nabla \underline{u} + \nabla \underline{u}^T)$$

$$\nabla \cdot \underline{\underline{\sigma}} = 0$$



Elasticity

- Stress σ , displacement u , Lamé coefficients λ, μ

$$\underline{\underline{\sigma}} = \lambda \nabla \cdot \underline{u} I + \mu (\nabla \underline{u} + \nabla \underline{u}^T)$$

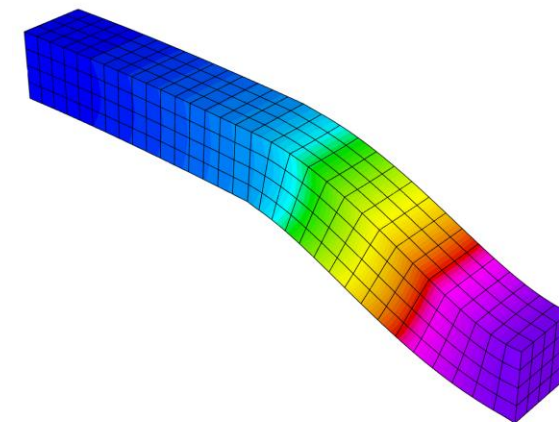
$$\nabla \cdot \underline{\underline{\sigma}} = 0$$

- Split form $\dim(\sigma_1)=1, \dim(\sigma_2)=d(d+1)/2$

$$\sigma_1 = \lambda \nabla \cdot \underline{u}$$

$$\sigma_2 = 2\mu \nabla_s \underline{u}$$

$$\nabla \sigma_1 + \nabla_s \cdot \sigma_2 = \underline{0}$$

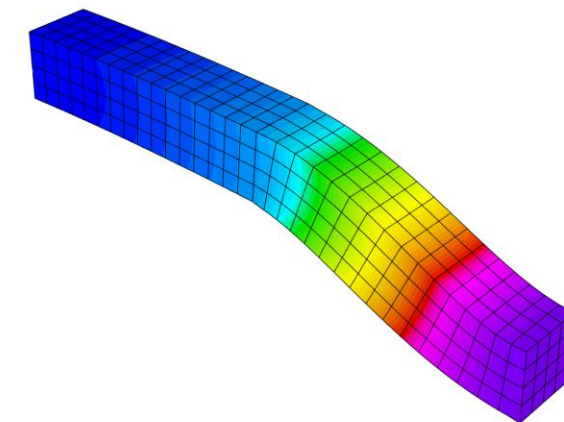


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$$\begin{pmatrix} \frac{1}{\lambda} & \\ & \frac{1}{2\mu} \end{pmatrix} \sigma - \begin{pmatrix} \nabla \cdot \\ \nabla_s \end{pmatrix} \underline{u} = \underline{0}$$

$$\begin{pmatrix} \nabla \\ \nabla_s \cdot \end{pmatrix}^T \sigma = \underline{0}$$

HDG elasticity

- Lagrange multiplier $\hat{u}_h$ (DG or H1 trace space)

$$\tau = \frac{\lambda + 2\mu}{h}$$

$$\begin{aligned} & \left(\left(\begin{pmatrix} \frac{1}{\lambda} & \\ & \frac{1}{2\mu} \end{pmatrix} \sigma_h, \xi \right) + \left(u_h, \begin{pmatrix} \nabla \\ \nabla_{s \cdot} \end{pmatrix} \xi \right) - \left\langle \hat{u}_h, \begin{pmatrix} \llbracket \xi_1 \rrbracket \underline{n} \\ \llbracket \xi_2 \cdot \underline{n} \rrbracket \end{pmatrix} \right\rangle = 0 \quad \forall \xi \\ & - \left(\left(\begin{pmatrix} \nabla \\ \nabla_{s \cdot} \end{pmatrix}^T \sigma_h, v \right) + \langle \tau u_h, v \rangle - \langle \tau \hat{u}_h, v \rangle = \underline{0} \quad \forall v \\ & - \left\langle \begin{pmatrix} \llbracket \sigma_{h1} \rrbracket \underline{n} \\ \llbracket \sigma_{h2} \cdot \underline{n} \rrbracket \end{pmatrix}^T, \hat{v} \right\rangle + \langle \tau u_h, \hat{v} \rangle - \langle \tau \hat{u}_h, \hat{v} \rangle = \underline{0} \quad \forall \hat{v} \end{aligned}$$

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- Smaller system for $\hat{u}_h$ only

$$\begin{bmatrix} \underline{A} & \underline{-B}^T & \underline{C}^T \\ \underline{B} & \underline{D} & \underline{E} \\ \underline{C} & \underline{G} & \underline{H} \end{bmatrix} \begin{bmatrix} \underline{Q} \\ \underline{U} \\ \underline{A} \end{bmatrix} = \begin{bmatrix} \underline{R} \\ \underline{F} \\ \underline{L} \end{bmatrix} \Rightarrow \boxed{\mathbb{K} \hat{u}_h = \mathbb{F}}$$

HDG elasticity

- Lagrange multiplier $\hat{u}_h$ (DG or H1 trace space)

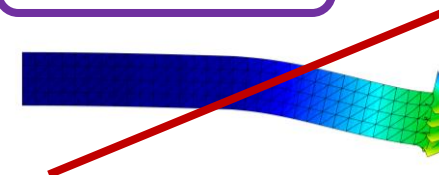
$$\tau = \frac{\lambda+2\mu}{h}$$

$$\begin{aligned} & \left(\left(\begin{array}{cc} \frac{1}{\lambda} & \\ & \frac{1}{2\mu} \end{array} \right) \sigma_h, \xi \right) + \left(u_h, \left(\begin{array}{c} \nabla \\ \nabla_{s \cdot} \end{array} \right) \xi \right) - \left\langle \hat{u}_h, \left(\begin{array}{c} \llbracket \xi_1 \rrbracket \underline{n} \\ \llbracket \xi_2 \cdot \underline{n} \rrbracket \end{array} \right) \right\rangle = 0 \quad \forall \xi \\ & - \left(\left(\begin{array}{c} \nabla \\ \nabla_{s \cdot} \end{array} \right)^T \sigma_h, v \right) + \langle \tau u_h, v \rangle - \langle \tau \hat{u}_h, v \rangle = 0 \quad \forall v \\ & - \left\langle \left(\begin{array}{c} \llbracket \sigma_{h1} \rrbracket \underline{n} \\ \llbracket \sigma_{h2} \cdot \underline{n} \rrbracket \end{array} \right)^T, \hat{v} \right\rangle + \langle \tau u_h, \hat{v} \rangle - \langle \tau \hat{u}_h, \hat{v} \rangle = 0 \quad \forall \hat{v} \end{aligned}$$

- Smaller system for $\hat{u}_h$ only

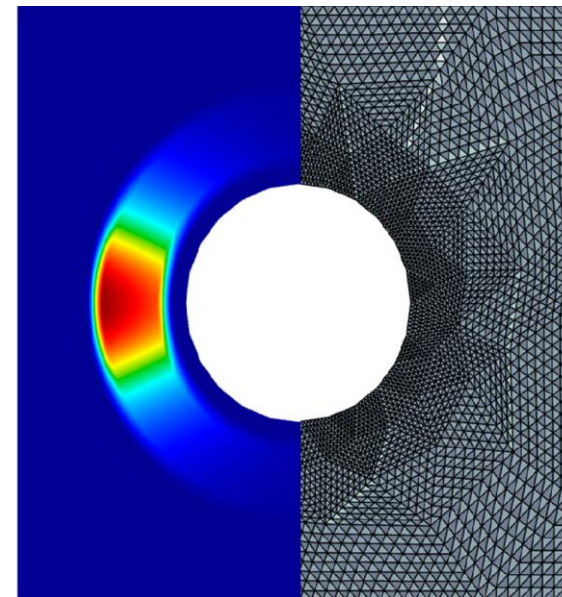
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- Dirichlet (displacement) BC – natural / essential



Anisotropic HDG h -adaptivity

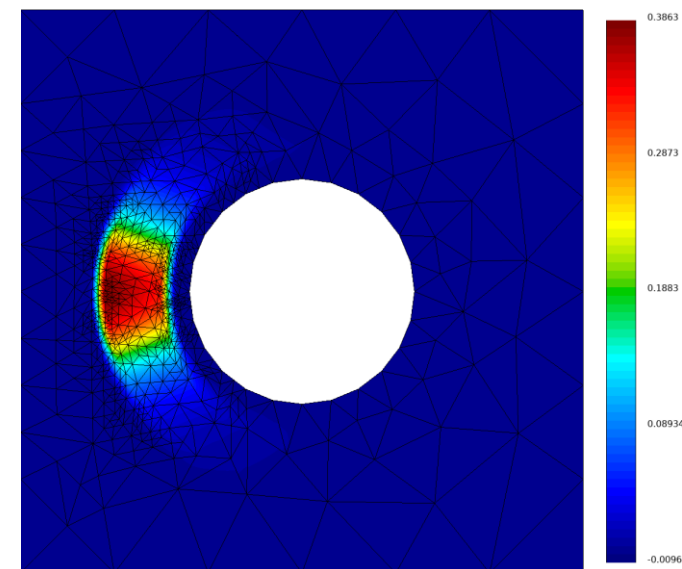
- Different scales ($\kappa_{\parallel} \gg \kappa_{\perp}$)
- Diffusion ring (10^8 anisotropy)
- Zienkiewicz-Zhu $\sim \vec{\kappa}(\nabla u_h - \overline{\nabla u_h})$



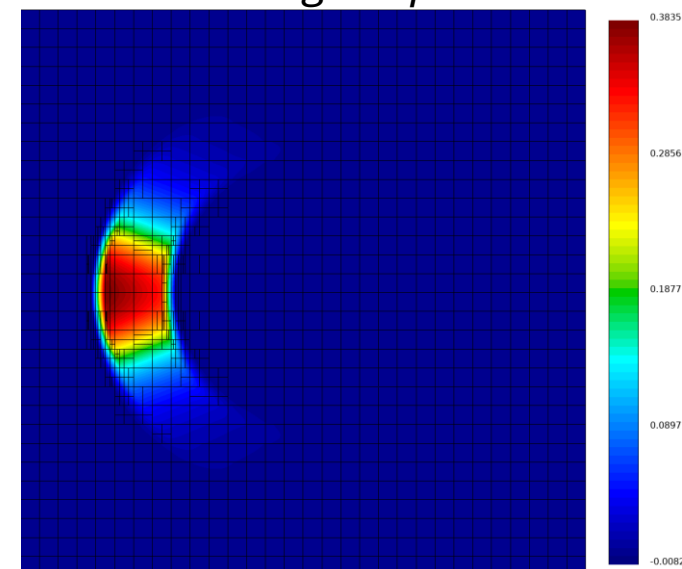
Anisotropic HDG h -adaptivity

- Different scales ($\kappa_{\parallel} \gg \kappa_{\perp}$)
- Diffusion ring (10^8 anisotropy)
- Zienkiewicz-Zhu $\sim \hat{\vec{\kappa}}(\nabla \mathbf{u}_h - \overline{\nabla \mathbf{u}_h})$
- HDGErrorEstimator

$$\langle \vec{n} \cdot \hat{\vec{\kappa}} \cdot \vec{n} (\lambda_h - \mathbf{u}_h)^2 / h \rangle$$
- Conforming x non-conforming



Conforming simplex AMR

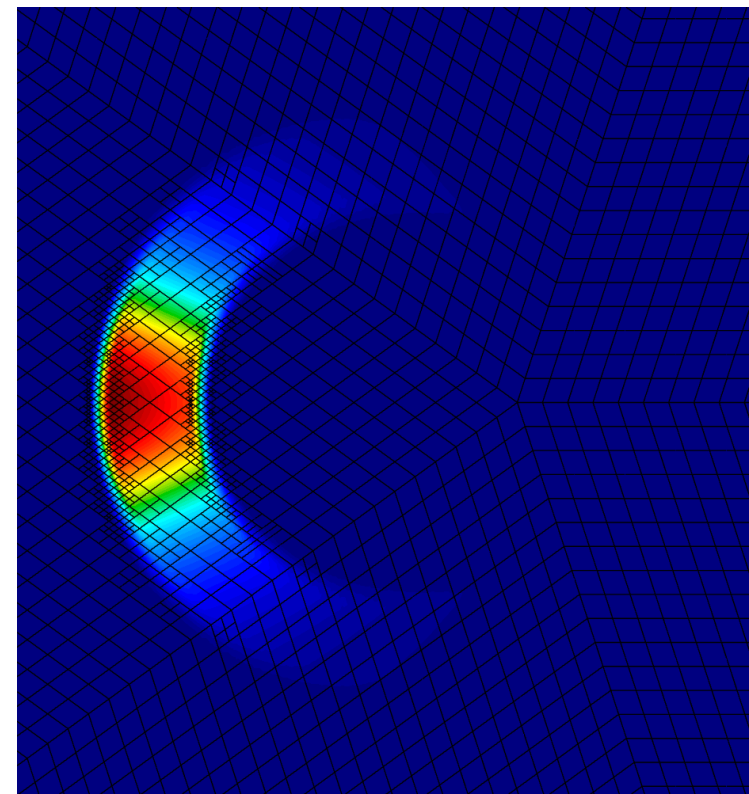


Non-conforming tensor AMR

Anisotropic HDG h -adaptivity

- Different scales ($\kappa_{\parallel} \gg \kappa_{\perp}$)
- Diffusion ring (10^8 anisotropy)
- Zienkiewicz-Zhu $\sim \vec{\kappa}(\nabla u_h - \overline{\nabla u_h})$
- HDGErrorEstimator

$$\langle \vec{n} \cdot \vec{\kappa} \cdot \vec{n} (\lambda_h - u_h)^2 / h \rangle$$
- Conforming x non-conforming
- Arbitrary meshes



Single null diverted tokamak

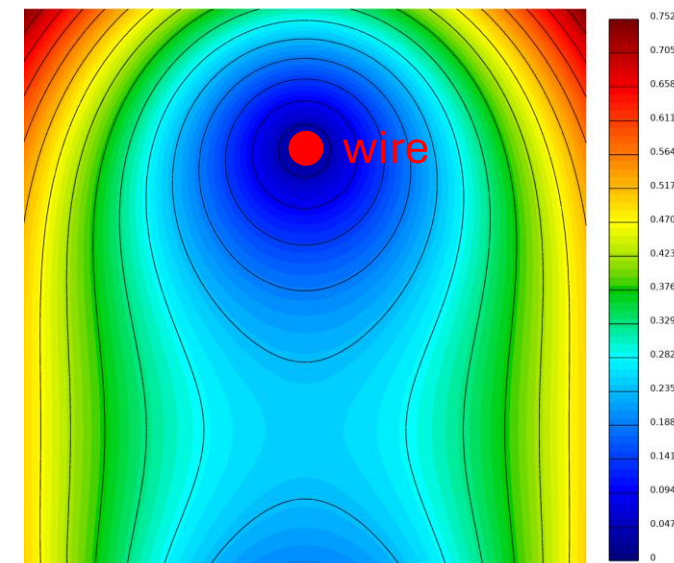
- Two-wire approximation

$$A_z = \log(|\vec{x} - \vec{x}_1| |\vec{x} - \vec{x}_2|)$$

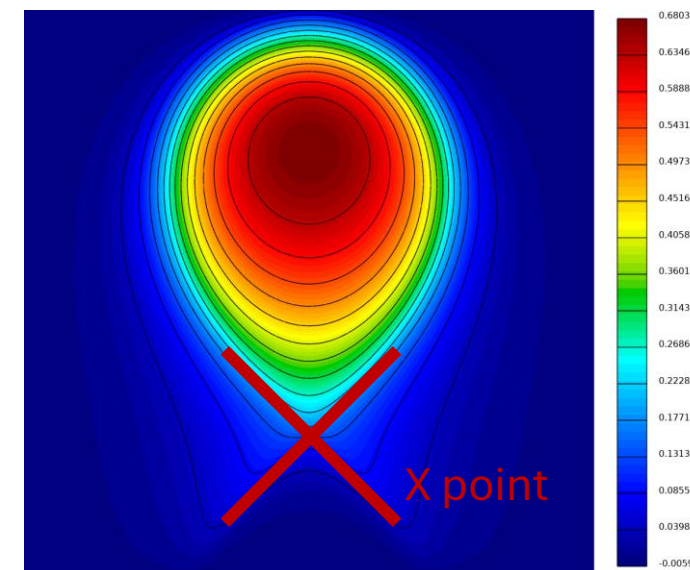
C. Vogl, I. Joseph, & M. Holec (2023). Mesh refinement for anisotropic diffusion in magnetized plasmas. *Comput. Math. Appl.*, 145, 159-174.

- Large aspect ratio approx.

$$\vec{B}_t = \vec{z} \quad \vec{B}_p = \nabla \times (A_z \vec{z})$$



Mag. potential (exp A_z)



Potential ($\kappa_{\parallel}/\kappa_{\perp} = 10^2$)

Single null diverted tokamak

- Two-wire approximation

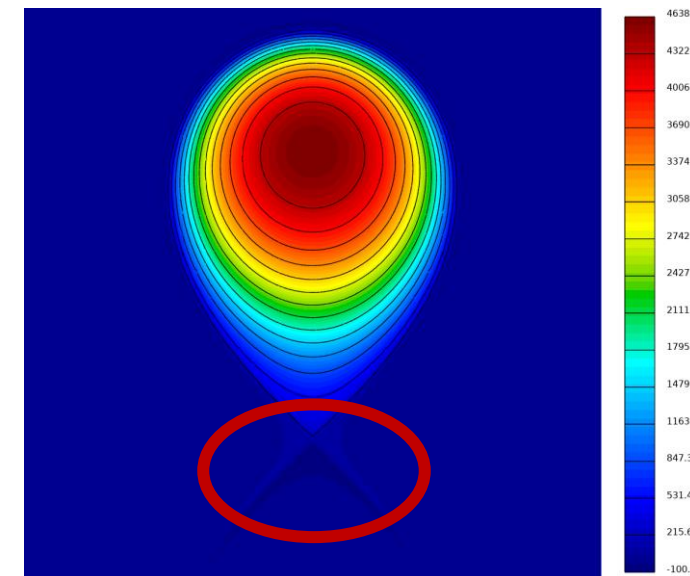
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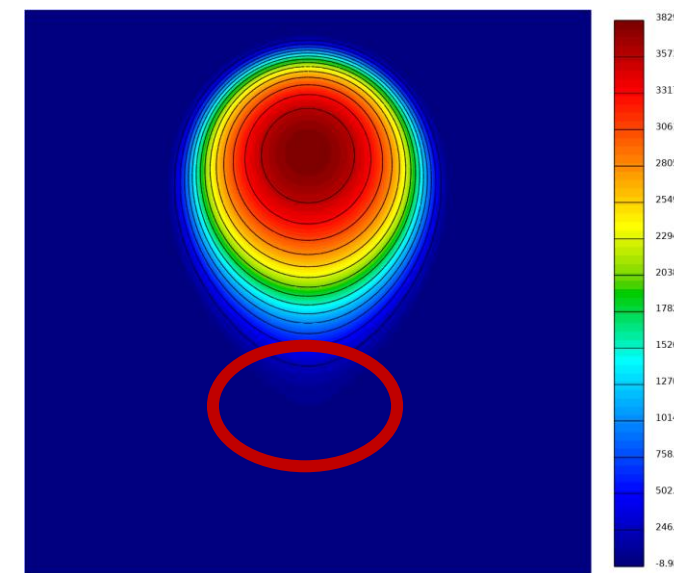
- Large aspect ratio approx.

$$\vec{B}_t = \vec{z} \quad \vec{B}_p = \nabla \times (A_z \vec{z})$$

- $\kappa_{\parallel} / \kappa_{\perp} = 10^6$, thickness $\sim \sqrt{\kappa_{\perp} / \kappa_{\parallel}}$



HRT



HDG

Single null diverted tokamak

- Two-wire approximation

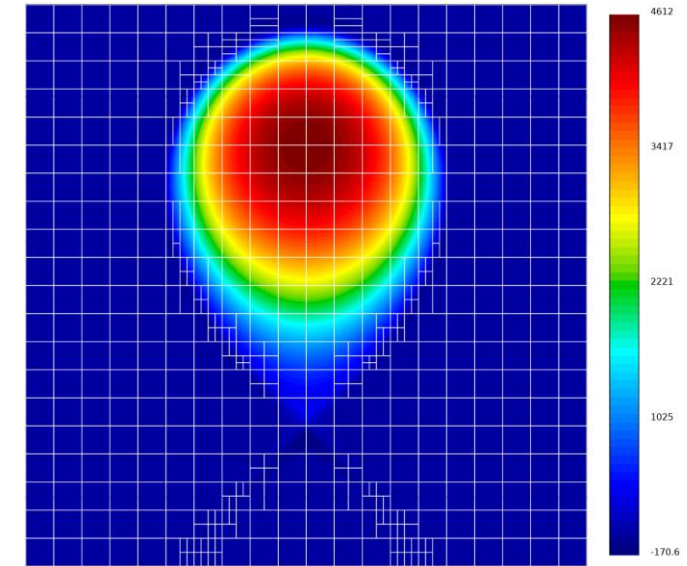
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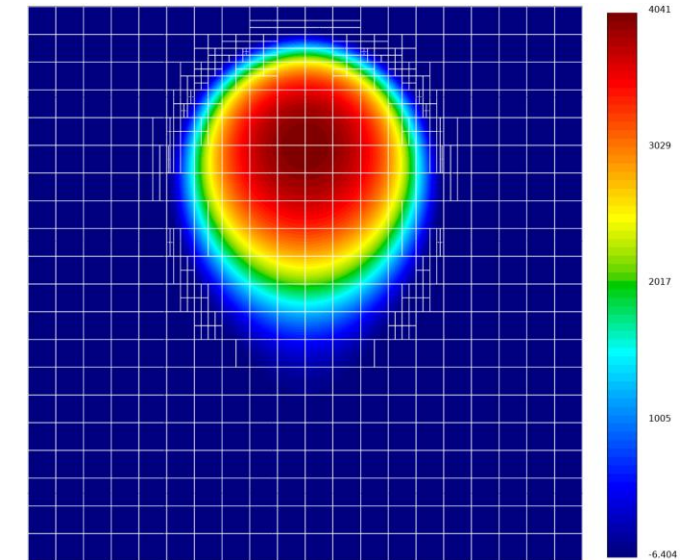
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HRT



HDG

Single null diverted tokamak

- Two-wire approximation

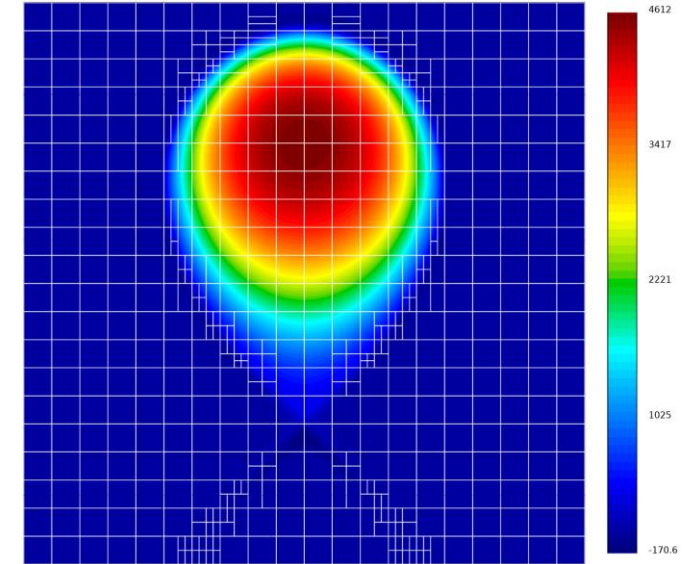
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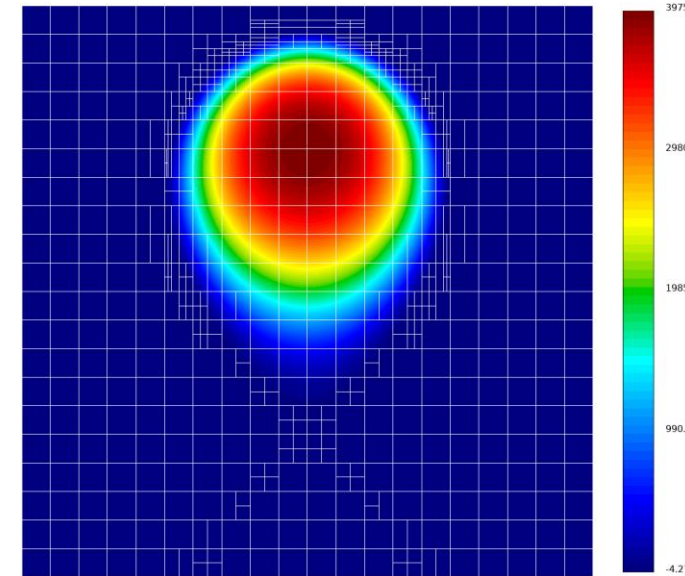
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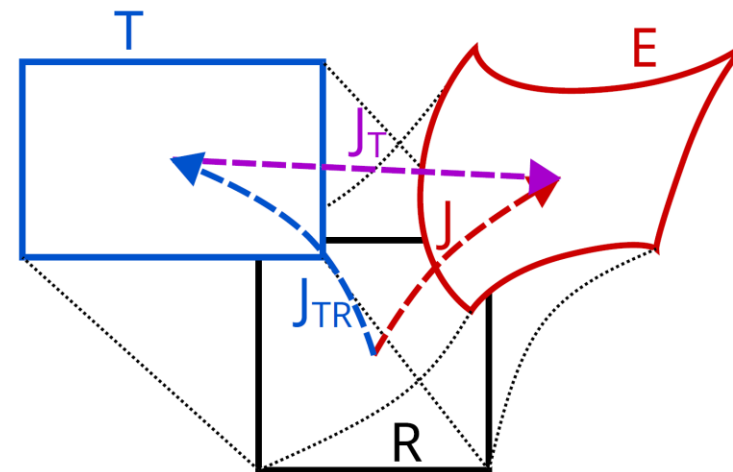


HDG

PDE-driven r -adaptivity

- Target-Matrix Optimization Paradigm (TMOP)

$$F_\mu = \sum \int_E \mu(J_T(\mathbf{x}^E))$$

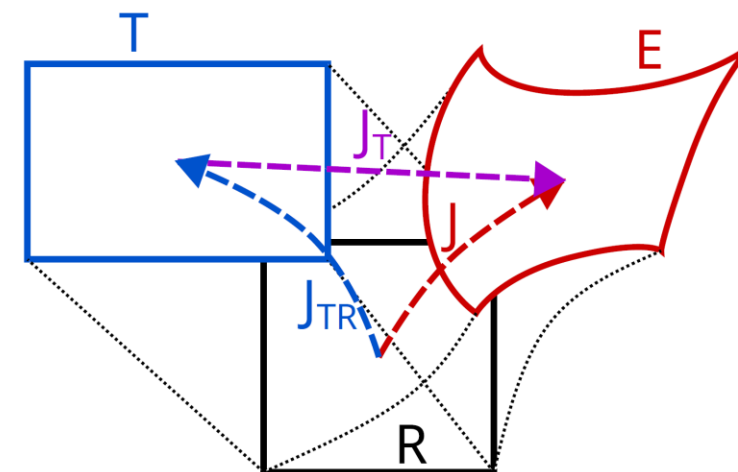


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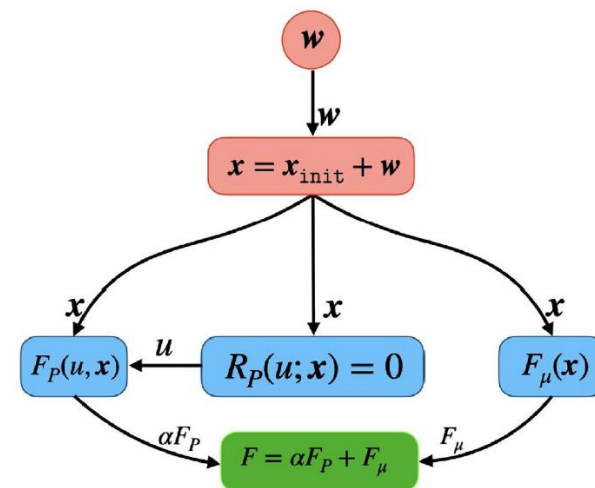
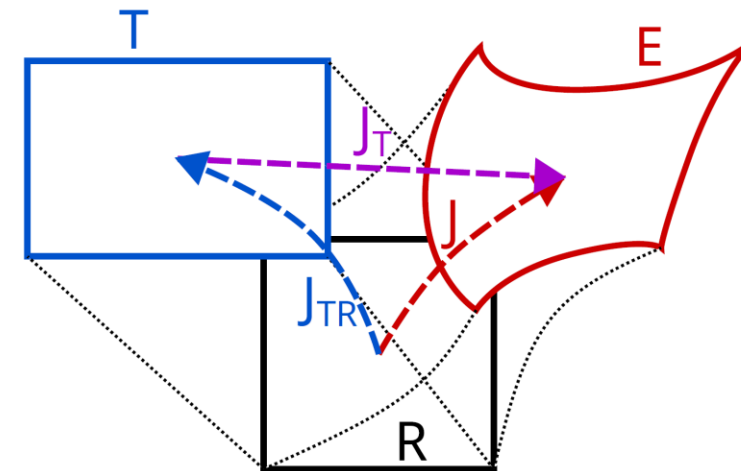
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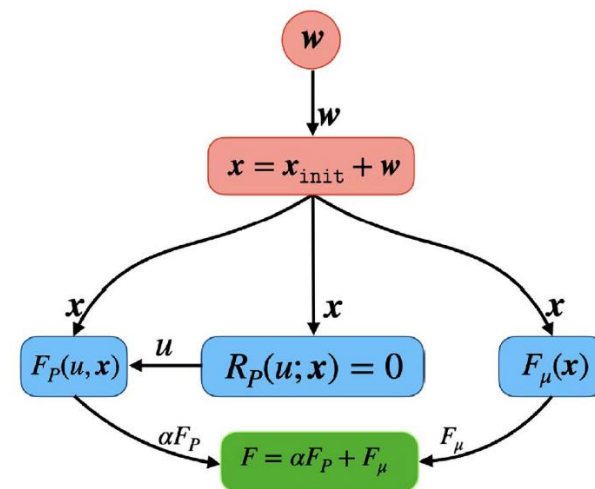
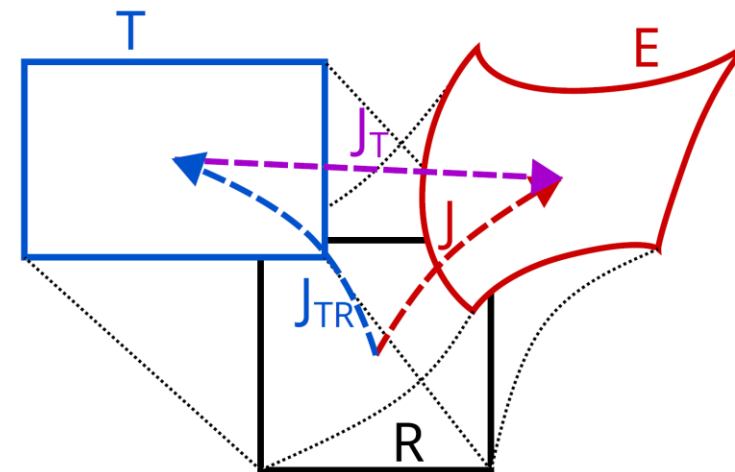
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- Newton iterations (for $\mathbf{w}$)



Anisotropic HDG r -adaptivity

- $F_P = \langle \vec{n} \cdot \vec{\kappa} \cdot \vec{n} (\lambda_h - u_h)^2 / h \rangle$

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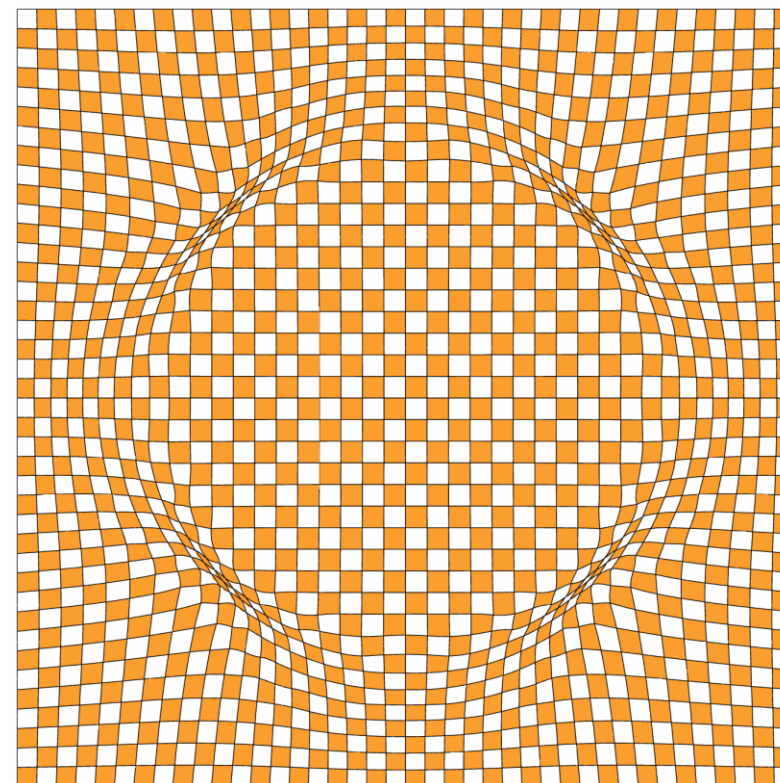
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Anisotropic HDG r -adaptivity

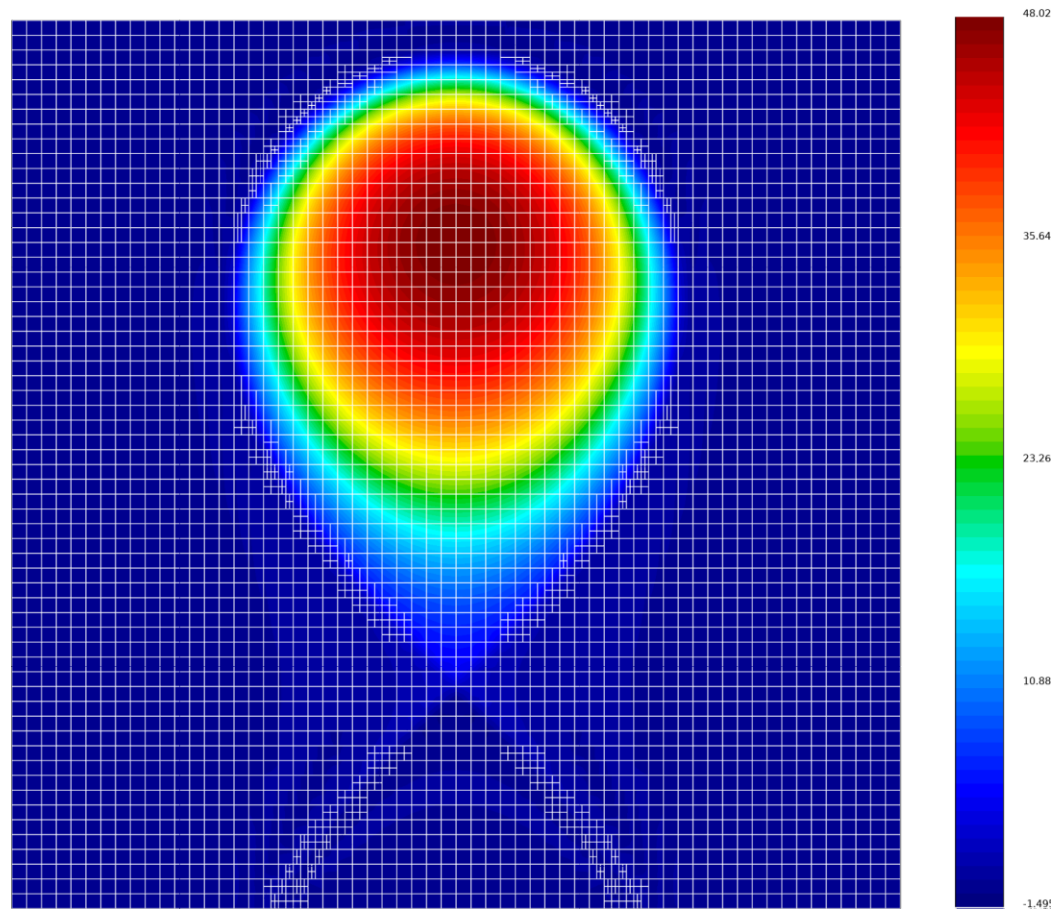
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- α based on energy ratio



Gaussian ring – energy ratio 0.5

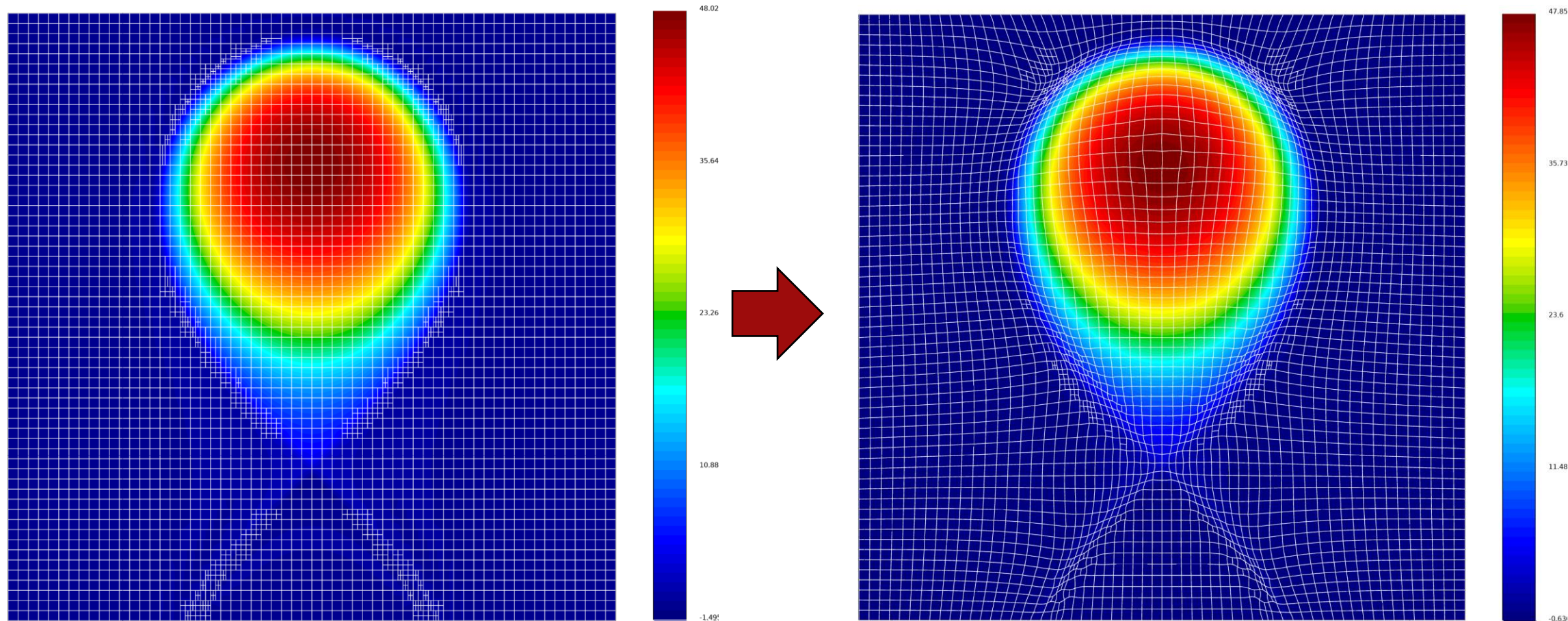
Anisotropic HDG *hr*-adaptivity



h-adaptation

$Q1$ 60x60 HRT $\kappa_{\parallel}/\kappa_{\perp} = 10^4$ - 5 iterations

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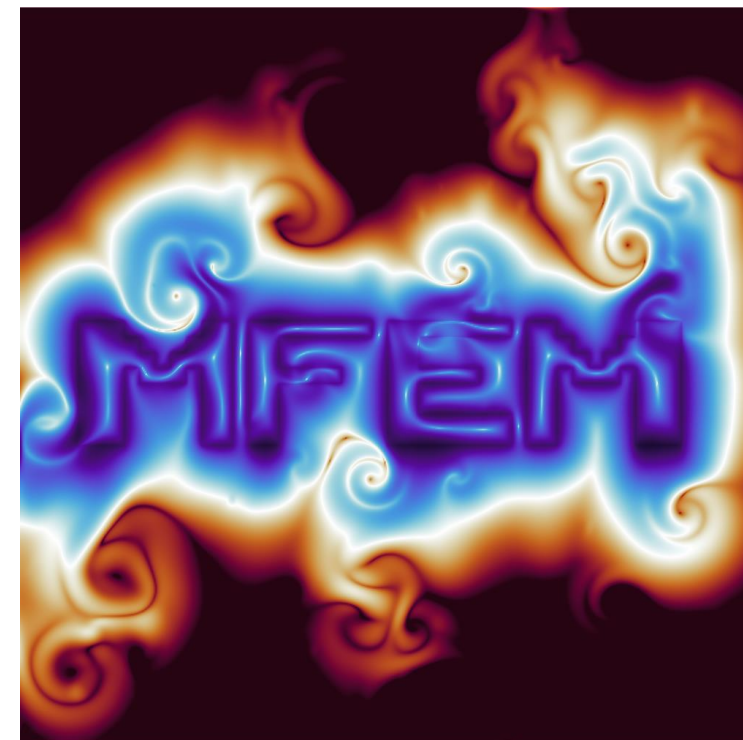
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hr-adaptation (0.1 energy ratio)

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Conclusions

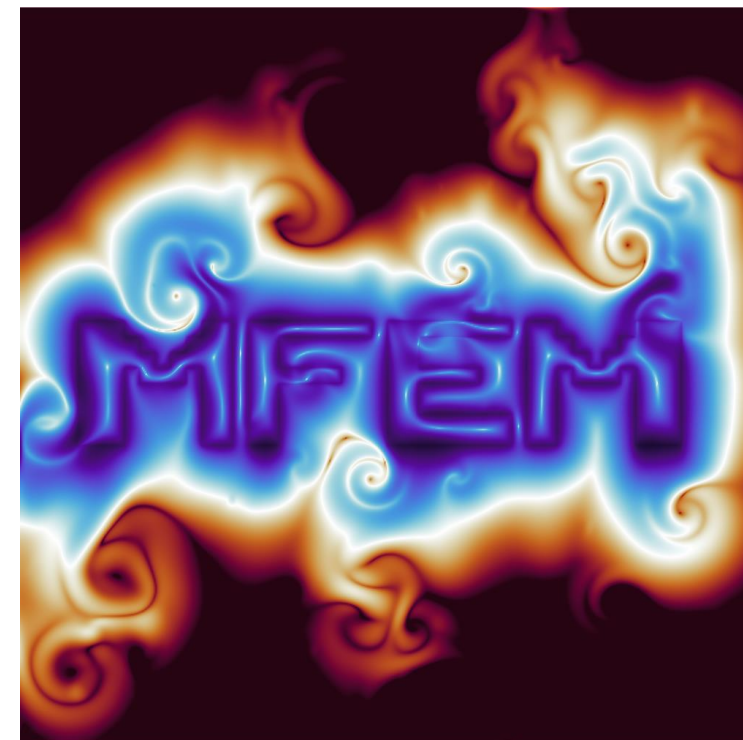
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Single-step anisotropic
diffusion-convection simulation
(mfem-logo)

Conclusions

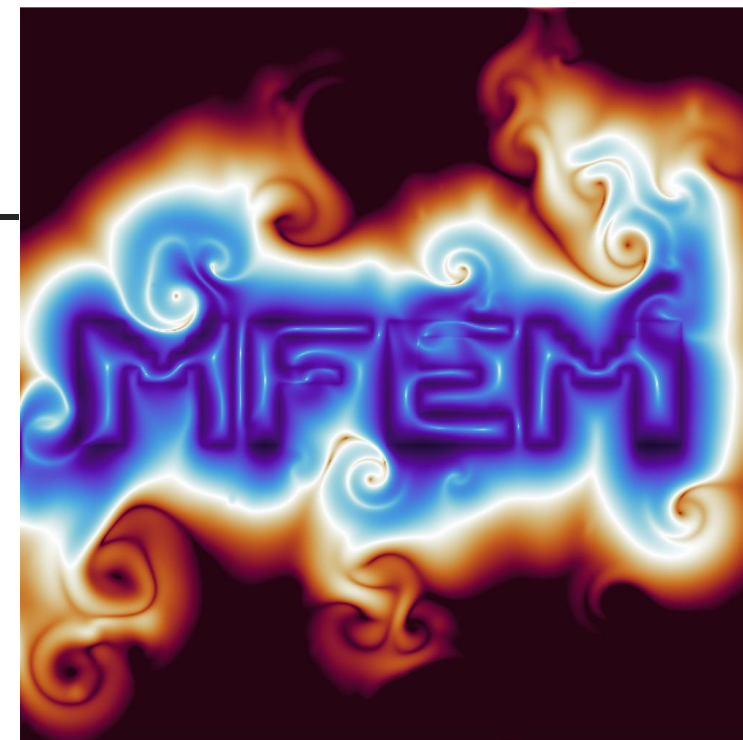
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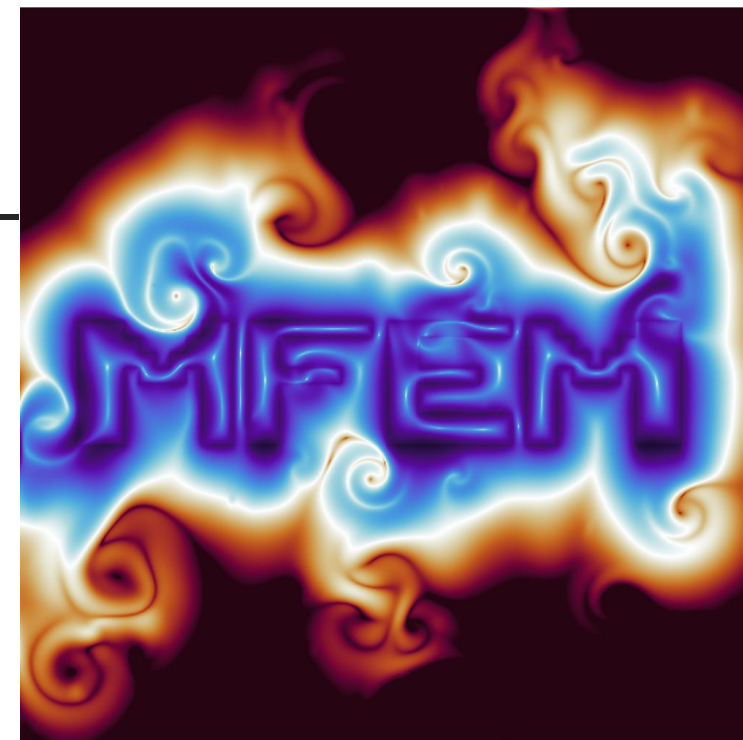
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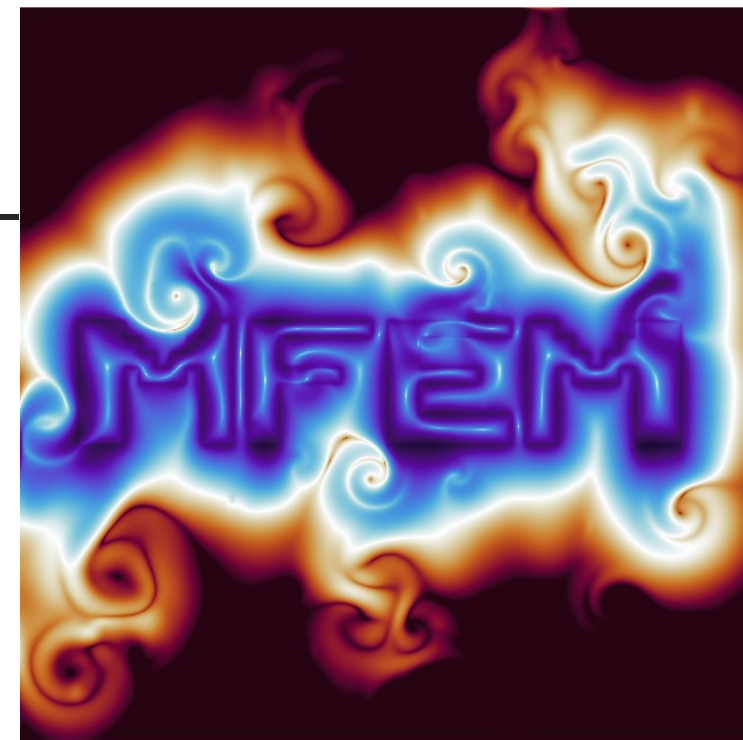
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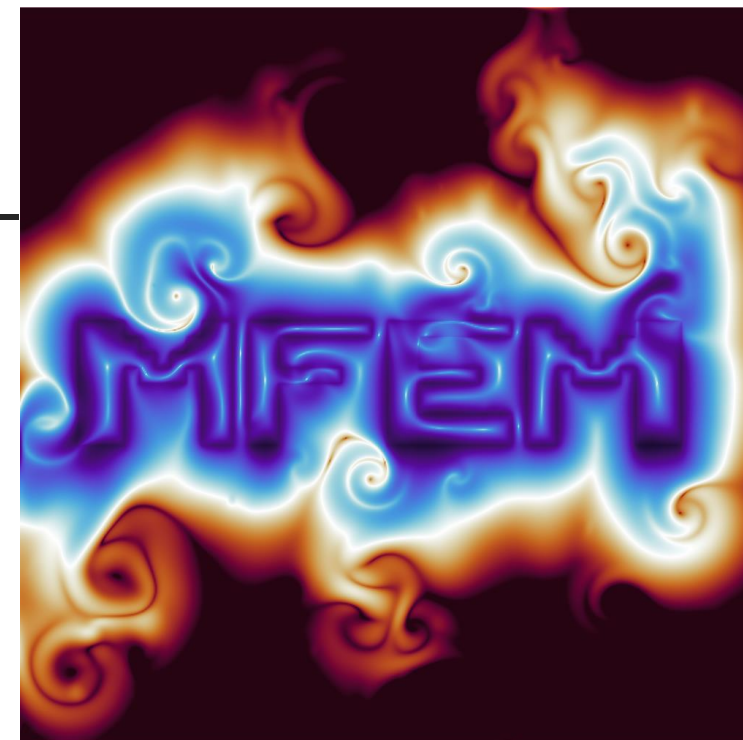
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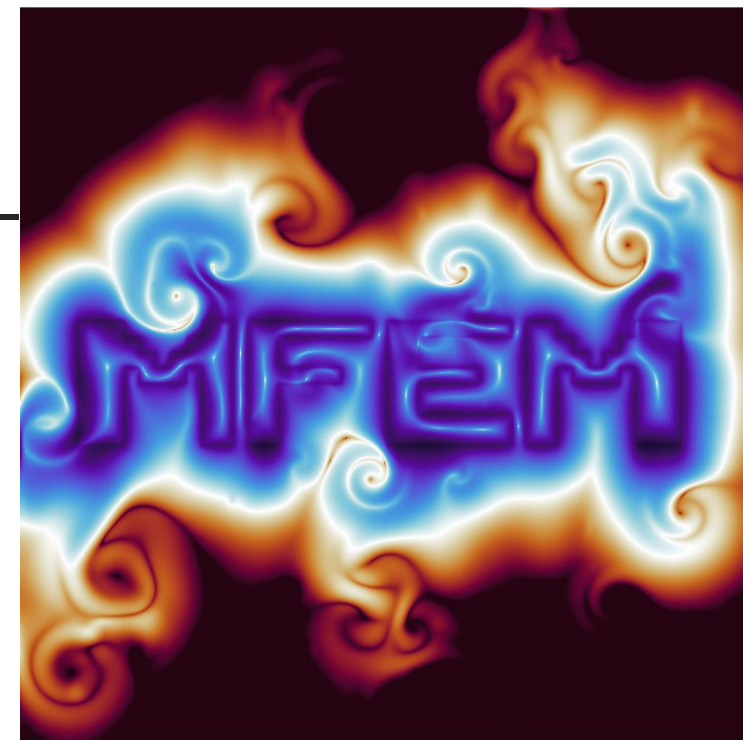
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- TODOs – algebraic constr., GPUs, Maxwell, ...



Single-step anisotropic
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(mfem-1.0.0)



Thank you for your attention

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